Odd Characterization and Transport Stack Note

Internal technical note

March 30, 2026

Abstract

This wrapper compiles the compressed welded LoG spectral-selection note, the raw odd-selector note, the canonical odd-interface note, and the active-response transport note. It serves as a stable ‘xr‘ target for downstream label resolution.

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1 Canonical Odd Spectral Selection on the Compressed Welded LoG Channel

Fix an accepted tick k on a welded admissible branch. This section isolates the compressed welded LoG channel and proves that the continued odd quadratic form canonically selects a negative rank-one odd line on that channel. The transport of the descendant class into that line is recorded separately in theorem 3.20; here it is used only in corollary 1.13.

The macro source-side temporal envelope is DoG, the macro source-side spatial envelope is LoG, the micro receiver temporal envelope is DoG, and the micro receiver spatial envelope is Gaussian. By Corollary 3.1, the welded envelope is LoG in both time and space, with scales adding in quadrature.

Definition 1.1 (Canonical welded LoG profile)**.** Let t_k be the accepted tick time and let $\Omega_* \in S^3$ denote the common spatial center selected by the co-centering/history-lock mechanism. Let

$$s_{t,k} := \sqrt{\Sigma_k^2 + \sigma_{t,k}^2}, \qquad s_{x,k} := \sqrt{\rho_k^2 + \sigma_{x,k}^2}$$

be the effective welded scales. Define the normalized welded LoG pulse by

$$u_{\Lambda_k}^{\mathrm{LoG}}(t,\Omega) := \mathcal{N}_k \frac{d^2}{dt^2} g_{s_{t,k}}(t-t_k)\, \Delta_{S^3} g_{s_{x,k}}(\Omega,\Omega_*), \tag{1}$$

where $\mathcal{N}_k > 0$ is chosen so that $\|u_{\Lambda_k}^{\mathrm{LoG}}\|_{L^2(\mathbb{R}\times S^3)} = 1$. We write

$$\mathcal{H}_{\Lambda_k}^{\mathrm{LoG}} := \mathbb{R}\, u_{\Lambda_k}^{\mathrm{LoG}} \tag{2}$$

for the compressed welded LoG channel.

Proposition 1.2 (Exact welded LoG form)**.** *The normalized profile $u^{\mathrm{LoG}}_{\Lambda_k}$ of definition 1.1 is exactly the welded macro/micro profile produced by:*

1. *macro temporal DoG convolved with micro temporal DoG;*
2. *macro spatial LoG convolved with micro spatial Gaussian.*

In particular the welded profile is LoG in time and LoG in space, with scales adding in quadrature.

Proof. The temporal and spatial welded identities are exactly the two statements of Corollary 3.1. After multiplying the factors and normalizing in L^2, one obtains (1). □

Proposition 1.3 (Band-edge locking and unique center)**.** *Assume the accepted tick lies on an admissible band-locked welded branch in the sense of Definition 3.2. Then:*

1. *the effective welded widths satisfy*

$$s_{t,k} = \frac{\alpha_t^*}{\Lambda_k}, \qquad s_{x,k} = \frac{\alpha_x^*}{v\,\Lambda_k},$$

   with $\alpha_t^*, \alpha_x^* > 0$ independent of k along a fixed admissible branch;

2. *the macro and micro temporal centers coincide at t_k, and the macro and micro spatial centers coincide at Ω_*;*
3. *once the receiver trace sector carries a nonzero inherited LoG background, the absolute center Ω_* is uniquely locked for all subsequent accepted ticks.*

Consequently, after normalization, the admissible welded pulse space at fixed accepted tick is the one-dimensional line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$.

Proof. The width lock is the band-locking clause of Definition 3.2. The temporal and spatial co-centering statements are ($\mathrm{C_t}$) and ($\mathrm{C_x}$), and the persistence of the spatial center is exactly Proposition 3.5.

Therefore, at fixed accepted tick, the shape, widths, and center are fixed. The only remaining freedom is an overall real amplitude. After L^2-normalization, that freedom reduces to the sign $\pm u^{\mathrm{LoG}}_{\Lambda_k}$. Hence the admissible welded channel is precisely the one-dimensional real line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. □

Proposition 1.4 (Smooth dependence along a branch)**.** *Along any admissible branch on which $\Lambda \mapsto \alpha_t^*, \alpha_x^*$ are constant and the locked center Ω_* is fixed, the map*

$$\Lambda \longmapsto u^{\mathrm{LoG}}_{\Lambda}$$

is C^1 in $L^2(\mathbb{R} \times S^3)$. Consequently the rank-one orthogonal projector

$$P^{\mathrm{LoG}}_{\Lambda} := u^{\mathrm{LoG}}_{\Lambda} \otimes u^{\mathrm{LoG}}_{\Lambda} \tag{3}$$

is C^1 along the branch.

Proof. Along the branch one has $s_t(\Lambda) = \alpha_t^*/\Lambda$ and $s_x(\Lambda) = \alpha_x^*/(v\Lambda)$, so both scales depend C^∞-smoothly on Λ. The Gaussian heat kernel and its time and spatial derivatives depend smoothly on the scale parameter, hence the unnormalized welded LoG profile depends C^1-smoothly on Λ in L^2. Its L^2-norm is strictly positive, so normalization preserves C^1-dependence. Formula (3) then gives a C^1 projector-valued map. □

Proposition 1.5 (Well-defined restriction of the continued quadratic form)**.** *Assume the welded linear-response/Gaussian radiative-sector package of Paragraph 3.5 and Theorem E.1. Then the analytically continued odd quadratic response form is well-defined on the compressed welded channel* $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$.

Proof. By Paragraph 3.5, the welded source enters the Gaussian radiative sector through the linear response map T_Λ, and Theorem E.1 supplies the finite-cutoff bounded Gaussian response package on the welded envelope class. By proposition 1.3, the compressed welded channel is a one-dimensional subspace of that admissible class. Restricting any continuous quadratic form to a one-dimensional subspace is automatic. Hence the analytically continued odd quadratic response form restricts canonically to $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. □

Definition 1.6 (Compressed continued odd quadratic coefficient)**.** Fix a witness window $\mu_{ps} > 0$ in the accepted regime. Let

$$\mathfrak{q}^{\mathrm{cont}}_{\Lambda_k,\mu_{ps}} : \mathcal{H}^{\mathrm{LoG}}_{\Lambda_k} \times \mathcal{H}^{\mathrm{LoG}}_{\Lambda_k} \to \mathbb{R}$$

denote the analytically continued odd quadratic response form restricted to the compressed welded LoG channel. Since $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$ is one-dimensional, there is a unique scalar $\beta_k \in \mathbb{R}$ such that

$$\mathfrak{q}^{\mathrm{cont}}_{\Lambda_k,\mu_{ps}}(X,Y) = \beta_k \, \langle X, u^{\mathrm{LoG}}_{\Lambda_k}\rangle_{L^2} \langle Y, u^{\mathrm{LoG}}_{\Lambda_k}\rangle_{L^2} \qquad \forall X, Y \in \mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}. \tag{4}$$

Equivalently,

$$\beta_k = \mathfrak{q}^{\mathrm{cont}}_{\Lambda_k,\mu_{ps}}\big(u^{\mathrm{LoG}}_{\Lambda_k}, u^{\mathrm{LoG}}_{\Lambda_k}\big).$$

Proposition 1.7 (Rigorous control of the compressed coefficient)**.** *Assume:*

1. *the smooth RP degradation law of Theorem D.1;*
2. *the approximate intertwiner bound of Lemma E.2;*
3. *the temporal DoG commutator bound of Lemma E.3;*
4. *the exponentially decaying memory-tail package of Lemma C.2 and Proposition C.1.*

Then there exist constants $C_{\mathrm{int}}, C_{\mathrm{DoG}}, C_{\mathrm{mem}}, C_4 > 0$ *and* $\eta > 0$*, depending only on the fixed welded family and not on* k*, such that*

$$\beta_k = -\,\alpha(\Lambda_k)\,\mu_{ps}^2 \;+\; r_k, \tag{5}$$

with remainder bounded by

$$|r_k| \le C_{\mathrm{int}}\,\varepsilon_k^{\eta} \;+\; C_{\mathrm{DoG}}\,\Phi(\sigma_{t,k}\Omega_{\mathrm{gap}}) \;+\; C_{\mathrm{mem}}\,e^{-\Delta_k/\tau_b} \;+\; C_4\,\mu_{ps}^4. \tag{6}$$

Proof. By Theorem D.1, the ideal analytically continued Gaussian OS problem gives $\lambda_{\min}(\mu_{ps},\Lambda_k) = -\alpha(\Lambda_k)\mu_{ps}^2 + O(\mu_{ps}^4)$ on the welded Gaussian reference kernel. Restricting that quadratic form to the canonical normalized welded LoG profile $u^{\mathrm{LoG}}_{\Lambda_k}$ produces the leading term $-\alpha(\Lambda_k)\mu_{ps}^2$, with the $O(\mu_{ps}^4)$ correction bounded by $C_4\mu_{ps}^4$.

The regulated intertwiner contribution is bounded by Lemma E.2, the temporal DoG contribution is bounded by Lemma E.3, and the intrinsic KMS bath contributes only the Born-Nakajima-Zwanzig memory tail controlled by Lemma C.2 and Proposition C.1. Since $u^{\mathrm{LoG}}_{\Lambda_k}$ is normalized, each of these operator bounds transfers directly to the scalar quadratic coefficient. Summing the four contributions gives (5)–(6). □

Definition 1.8 (Certified compressed spectral regime)**.** We say that an accepted tick k lies in the *certified compressed spectral regime* if

$$\alpha(\Lambda_k)\,\mu_{ps}^2 > 2\Big(C_{\mathrm{int}}\,\varepsilon_k^{\eta} \;+\; C_{\mathrm{DoG}}\,\Phi(\sigma_{t,k}\Omega_{\mathrm{gap}}) \;+\; C_{\mathrm{mem}}\,e^{-\Delta_k/\tau_b} \;+\; C_4\,\mu_{ps}^4\Big). \tag{7}$$

Theorem 1.9 (Finite-cutoff canonical odd spectral selection on the compressed welded LoG channel)**.** *Assume propositions 1.2 to 1.5 and 1.7. Let k be an accepted tick in the certified compressed spectral regime of definition 1.8. Then:*

1. *the compressed welded channel $\mathcal{H}_{\Lambda_k}^{\mathrm{LoG}}$ is a canonical one-dimensional odd line;*
2. *the compressed continued odd quadratic form is represented by the rank-one self-adjoint operator*

$$K_{\Lambda_k}^{\mathrm{LoG}} := \beta_k\,P_{\Lambda_k}^{\mathrm{LoG}}, \tag{8}$$

   with $\beta_k < 0$;
3. *the unique negative spectral projector of $K_{\Lambda_k}^{\mathrm{LoG}}$ is*

$$P_{-,\Lambda_k}^{\mathrm{LoG}} = P_{\Lambda_k}^{\mathrm{LoG}} = u_{\Lambda_k}^{\mathrm{LoG}} \otimes u_{\Lambda_k}^{\mathrm{LoG}}; \tag{9}$$

4. *along any admissible branch, the projector $P_{-,\Lambda}^{\mathrm{LoG}}$ is C^1, hence the compressed active odd line is canonically and smoothly transported.*

Proof. By proposition 1.3, the admissible welded channel at fixed accepted tick is exactly the one-dimensional line $\mathcal{H}_{\Lambda_k}^{\mathrm{LoG}}$. By definition 1.6, the compressed continued odd quadratic form on that line is represented by the scalar β_k, equivalently by the self-adjoint rank-one operator (8).

By proposition 1.7,

$$\beta_k \le -\,\alpha(\Lambda_k)\mu_{ps}^2 \;+\; \Big(C_{\mathrm{int}}\,\varepsilon_k^{\eta} \;+\; C_{\mathrm{DoG}}\,\Phi(\sigma_{t,k}\Omega_{\mathrm{gap}}) \;+\; C_{\mathrm{mem}}\,e^{-\Delta_k/\tau_b} \;+\; C_4\,\mu_{ps}^4\Big).$$

Under the certified regime inequality (7), the bracketed term is strictly smaller than $\frac{1}{2}\alpha(\Lambda_k)\mu_{ps}^2$. Therefore

$$\beta_k < -\frac{1}{2}\,\alpha(\Lambda_k)\mu_{ps}^2 < 0.$$

Thus $K_{\Lambda_k}^{\mathrm{LoG}}$ has exactly one negative eigenvalue, namely β_k, and its unique negative spectral subspace is the entire line $\mathcal{H}_{\Lambda_k}^{\mathrm{LoG}}$. Since $P_{\Lambda_k}^{\mathrm{LoG}} = u_{\Lambda_k}^{\mathrm{LoG}} \otimes u_{\Lambda_k}^{\mathrm{LoG}}$, the negative spectral projector is exactly (9).

Finally, proposition 1.4 gives C^1-dependence of $\Lambda \mapsto u_{\Lambda}^{\mathrm{LoG}}$, hence also of $\Lambda \mapsto P_{-,\Lambda}^{\mathrm{LoG}}$. This proves the canonical branchwise transport of the compressed active odd line. □

Corollary 1.10 (Canonical compressed odd projector)**.** *For each accepted tick in the certified compressed spectral regime, the compressed, co-centered, band-locked welded LoG channel carries the canonical rank-one projector*

$$P_{-,\Lambda_k}^{\mathrm{LoG}}$$

onto its negative odd line. Equivalently,

$$\mathcal{H}_{\Lambda_k}^{\mathrm{LoG}} = \operatorname{ran} P_{-,\Lambda_k}^{\mathrm{LoG}}.$$

Proof. This is exactly theorem 1.9: the theorem identifies the negative spectral projector with the rank-one projector $u_{\Lambda_k}^{\mathrm{LoG}} \otimes u_{\Lambda_k}^{\mathrm{LoG}}$, whose range is $\mathcal{H}_{\Lambda_k}^{\mathrm{LoG}}$. □

Corollary 1.11 (Completion-sector realization of the compressed projector)**.** *Assume assumption 1.3. Then, for each accepted tick in the certified compressed spectral regime, any Gaussian completion realization identifies the compressed projector $P^{\mathrm{LoG}}_{-,\Lambda_k}$ with a unique rank-one active odd line in the completion sector. Equivalently, there exists a unit vector*

$$u_-(\Lambda_k)$$

spanning that realized line such that

$$P^{\mathrm{LoG}}_{-,\Lambda_k} = u_-(\Lambda_k) \otimes u_-(\Lambda_k)$$

in the chosen realization.

Proof. Under assumption 1.3, the finite active completion-mode sector is defined only up to symplectic equivalence. A one-dimensional real line is invariant under such equivalences as a subspace, even though a basis vector on that line is not canonical before normalization. By theorem 1.9, the compressed odd line is already canonically fixed abstractly by the projector $P^{\mathrm{LoG}}_{-,\Lambda_k}$. Choosing the unique unit vector on that line in a given realization yields the stated formula, and changing the completion realization changes only the coordinate representative, not the underlying rank-one line. □

Remark 1.12 (Scope of the compressed spectral theorem)*.* Theorem theorem 1.9 determines the odd line only on the compressed welded LoG channel. It does not classify the full completion-sector odd subspace, and the completion-side statement of corollary 1.11 is only a realization statement for this rank-one projector.

Corollary 1.13 (Nontrivial descendant hit of the compressed dominant line)**.** *Assume theorem 3.20. Then for each accepted tick k, there exists a representative*

$$J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

whose response has nonzero component on the canonically selected compressed dominant odd line. Equivalently, the transported descendant sector maps nontrivially into the compressed line fixed by $P^{\mathrm{LoG}}_{-,\Lambda_k}$.

Proof. This is exactly item (1) of theorem 3.20. □

2 Raw Odd Selector and Scheme-Stable Affine Output

This section isolates the raw selector step on the odd descendant side of the UOM-to-Main interface used in Definition 4.2. At fixed finite cutoff, it records the Γ-odd affine output determined by a chosen sufficiently local anomaly representative and its same-scheme and inter-scheme translation laws.

Convention on the two $\mathbb{Z}_2$ gradings. The cohomological degree for the Q_t-complex is written with bars:

$$(\cdot)_{\bar{0}}, \qquad (\cdot)_{\bar{1}}.$$

Physical parity refers to the swap involution Γ, with eigenspaces

$$\Gamma F = \pm F.$$

Thus the selector output constructed below is cohomologically degree 0 but physically odd. Unless bar-degree notation is used explicitly, "odd" below means Γ-odd rather than Q_t-degree 1.

Work at one accepted tick k and one branch value u at cutoff Λ_k. Use the raw UOM descendant package of theorems 6.9 and 6.11. In particular:

1. a sufficiently local good monotone representative
$$\mathcal{O}^{(0)}_{\Lambda_k}(u);$$
2. its canonical anomaly representative
$$\mathcal{A}^{(1)}_{\Lambda_k}(u) := Q^{\mathrm{phys}}_{u,\Lambda_k}\mathcal{O}^{(0)}_{\Lambda_k}(u);$$
3. a swap involution
$$\Gamma : \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{\imath}} \to \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{\imath}} \qquad (\bar{\imath} \in \{\bar{0}, \bar{1}\})$$
preserving sufficient locality and cohomological degree, and commuting with Q_t:
$$\Gamma Q_t = Q_t \Gamma;$$
4. swap-evenness of the canonical core,
$$\Gamma\mathcal{O}^{(0)}_{\Lambda_k}(u) = \mathcal{O}^{(0)}_{\Lambda_k}(u), \qquad \Gamma\mathcal{A}^{(1)}_{\Lambda_k}(u) = \mathcal{A}^{(1)}_{\Lambda_k}(u);$$
5. swap covariance of the fixed-scheme physical radial differential on the chosen monotone sector,
$$\Gamma Q^{\mathrm{phys}}_{u,\Lambda_k} = Q^{\mathrm{phys}}_{u,\Lambda_k}\Gamma;$$

Items (1) and (2) are contained in theorem 6.9; item (3) uses definition 6.1 and lemma 6.2; item (4) uses proposition 6.4; and item (5) uses proposition 6.3.

Fix a chosen Γ-stable selector class of sufficiently local odd representatives of the same Q_t-cohomology class as $\mathcal{A}^{(1)}_{\Lambda_k}(u)$, and restrict to representatives for which the exact descent equation

$$Q^{\mathrm{phys}}_{u,\Lambda_k}\mathcal{O}^{(0)}_{\Lambda_k}(u) + Q_t\widetilde{\Xi}^{(1)}_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \tag{10}$$

admits at least one sufficiently local even solution $\widetilde{\Xi}^{(1)}_{\Lambda_k}(u) \in \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}}$.

Definition 2.1 (Selector class and canonical odd kernel channel)**.** Let

$$\mathcal{C}_{\mathrm{sel},\Lambda_k}(u) \subset \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{1}}$$

be the chosen Γ-stable class of sufficiently local odd representatives of the same degree-one Q_t-cohomology class as $\mathcal{A}^{(1)}_{\Lambda_k}(u)$ for which (10) admits a sufficiently local even solution. For $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$, define its swap-even and swap-odd parts by

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,\pm}(u) := \frac{1}{2}\Big(\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \pm \Gamma\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u)\Big).$$

Define the canonical odd kernel channel by

$$\mathcal{K}^{-}_{\Lambda_k} := \ker(Q_t) \cap \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}} \cap \ker(\Gamma + \mathrm{id}).$$

Proposition 2.2 (Odd defect of a selector representative is always Q_t-exact)**.** *Let* $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$*. Then:*

1. $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u)$ *and* $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$ *are sufficiently local;*
2. $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u)$ *is a swap-even representative of the same* Q_t*-cohomology class as* $\mathcal{A}^{(1)}_{\Lambda_k}(u)$*;*
3. $$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) \in \mathrm{im}(Q_t) \cap \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{1}} \cap \ker(\Gamma + \mathrm{id}).$$

Proof. Sufficient locality is preserved by linear combinations, proving item (1).

Because $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u)$ and $\mathcal{A}^{(1)}_{\Lambda_k}(u)$ represent the same Q_t-cohomology class, there exists $Y_{\Lambda_k}(u) \in \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}}$ such that

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) - \mathcal{A}^{(1)}_{\Lambda_k}(u) = Q_t Y_{\Lambda_k}(u).$$

Taking the swap-even part and using $\Gamma Q_t = Q_t \Gamma$ and $\Gamma \mathcal{A}^{(1)}_{\Lambda_k}(u) = \mathcal{A}^{(1)}_{\Lambda_k}(u)$, we obtain

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u) - \mathcal{A}^{(1)}_{\Lambda_k}(u) = Q_t Y_{\Lambda_k,+}(u),$$

so item (2) follows.

Taking the swap-odd part of the same identity gives

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = Q_t Y_{\Lambda_k,-}(u).$$

By construction, $\Gamma \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = -\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$, which proves item (3). □

Definition 2.3 (Selected odd descendant affine class)**.** For $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$, define the associated raw odd selector output by

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big) := \Big\{ \Xi^-_{\Lambda_k}(u) \in \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}} \;\Big|\; \Gamma \Xi^-_{\Lambda_k}(u) = -\Xi^-_{\Lambda_k}(u), \quad Q_t \Xi^-_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) \Big\}.$$

Remark 2.4 (Dependence only on the swap-odd defect)*.* By theorem 2.5, the affine class $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$ depends only on the swap-odd defect $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$.

Theorem 2.5 (Selector decomposition into even representative channel and odd output)**.** *Let* $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$*, and let* $\widetilde{\Xi}^{(1)}_{\Lambda_k}(u)$ *be any sufficiently local even solution of* (10)*. Define*

$$\widetilde{\Xi}^{(1)}_{\Lambda_k,\pm}(u) := \frac{1}{2}\Big(\widetilde{\Xi}^{(1)}_{\Lambda_k}(u) \pm \Gamma \widetilde{\Xi}^{(1)}_{\Lambda_k}(u)\Big).$$

Then:

1. *the swap-even and swap-odd channels decouple as*
$$Q^{\mathrm{phys}}_{u,\Lambda_k} \mathcal{O}^{(0)}_{\Lambda_k}(u) + Q_t \widetilde{\Xi}^{(1)}_{\Lambda_k,+}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u), \tag{11}$$
$$Q_t \widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u); \tag{12}$$
2. *the raw odd selector output* $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$ *contains the odd projection of any full solution; in particular it is nonempty;*

3. *it is an affine space over the canonical odd kernel channel:*

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big) = \widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u) + \mathcal{K}^{-}_{\Lambda_k}.$$

Proof. Apply Γ to (10). Using $\Gamma Q^{\mathrm{phys}}_{u,\Lambda_k} = Q^{\mathrm{phys}}_{u,\Lambda_k}\Gamma$ on the chosen monotone sector and $\Gamma\mathcal{O}^{(0)}_{\Lambda_k}(u) = \mathcal{O}^{(0)}_{\Lambda_k}(u)$, we obtain

$$Q^{\mathrm{phys}}_{u,\Lambda_k}\mathcal{O}^{(0)}_{\Lambda_k}(u) + Q_t\Gamma\widetilde{\Xi}^{(1)}_{\Lambda_k}(u) = \Gamma\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u).$$

Adding and subtracting this equation from (10) yields (11) and (12), proving item (1).

Item (2) is immediate from item (1): the odd projection $\widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u)$ of any full solution belongs to the set in definition 2.3.

Now let $\Xi^{-}_{\Lambda_k,1}(u)$ and $\Xi^{-}_{\Lambda_k,2}(u)$ lie in $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$. Subtracting their defining equations gives

$$Q_t\big(\Xi^{-}_{\Lambda_k,1}(u) - \Xi^{-}_{\Lambda_k,2}(u)\big) = 0.$$

Because both are Γ-odd, their difference lies in $\mathcal{K}^{-}_{\Lambda_k}$. Conversely, if $K^{-}_{\Lambda_k} \in \mathcal{K}^{-}_{\Lambda_k}$, then

$$Q_t\big(\widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u) + K^{-}_{\Lambda_k}\big) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u),$$

and the sum remains Γ-odd. This proves item (3). □

Corollary 2.6 (The two selector channels)**.** *At fixed cutoff, the selector problem splits into exactly two channels.*

1. ***Kernel channel.*** *If $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = 0$, then*

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big) = \mathcal{K}^{-}_{\Lambda_k}.$$

   In particular, the canonical anomaly representative $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) = \mathcal{A}^{(1)}_{\Lambda_k}(u)$ produces only the canonical odd kernel sector.

2. ***Representative channel.*** *If $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) \neq 0$, then the raw odd selector output is the translated affine space*

$$\widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u) + \mathcal{K}^{-}_{\Lambda_k}$$

   solving

$$Q_t\Xi^{-}_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u).$$

Thus the selector output is determined by the canonical odd kernel channel together with the swap-odd part of the chosen anomaly representative.

Proof. This is an immediate specialization of theorem 2.5. □

Remark 2.7 (Kernel activity versus representative-driven activity). By corollary 2.6, the selector output is either the canonical odd kernel channel $\mathcal{K}^{-}_{\Lambda_k}$ or a translate of that channel by a particular solution of $Q_t\Xi^{-}_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$.

Proposition 2.8 (Same-scheme affine translation of selector outputs)**.** *Let*

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u), \qquad \widehat{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$$

be selector representatives in the same finite-cutoff scheme, and suppose there exists an even sufficiently local cochain $Y_{\Lambda_k}(u)$ such that

$$\widehat{\mathcal{A}}^{(1)}_{\Lambda_k}(u) - \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) = Q_t Y_{\Lambda_k}(u). \tag{13}$$

Define

$$Y^-_{\Lambda_k}(u) := \frac{1}{2}\Big(Y_{\Lambda_k}(u) - \Gamma Y_{\Lambda_k}(u)\Big).$$

Then

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widehat{\mathcal{A}}^{(1)}_{\Lambda_k}\big) = \mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big) + Y^-_{\Lambda_k}(u).$$

If $Y'_{\Lambda_k}(u)$ is another primitive satisfying (13), then

$$Y'^-_{\Lambda_k}(u) - Y^-_{\Lambda_k}(u) \in \mathcal{K}^-_{\Lambda_k}.$$

In particular, if $\widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$, then

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widehat{\mathcal{A}}^{(1)}_{\Lambda_k}\big) = \mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big).$$

Proof. Taking swap-odd parts of (13) gives

$$\widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) - \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = Q_t Y^-_{\Lambda_k}(u).$$

Let $\Xi^-_{\Lambda_k}(u) \in \mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$. Then

$$Q_t \Xi^-_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u),$$

so

$$Q_t\big(\Xi^-_{\Lambda_k}(u) + Y^-_{\Lambda_k}(u)\big) = \widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u).$$

Because both terms are Γ-odd, the sum lies in $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widehat{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$. This proves one inclusion, and the reverse inclusion follows by symmetry with $-Y^-_{\Lambda_k}(u)$.

If $Y'_{\Lambda_k}(u)$ is another primitive, then $Q_t(Y'_{\Lambda_k}(u) - Y_{\Lambda_k}(u)) = 0$. Taking swap-odd parts yields

$$Q_t\big(Y'^-_{\Lambda_k}(u) - Y^-_{\Lambda_k}(u)\big) = 0.$$

Since the difference is also Γ-odd, it lies in $\mathcal{K}^-_{\Lambda_k}$. If $\widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$, then one may choose $Y^-_{\Lambda_k}(u) = 0$, and the last claim follows. □

Theorem 2.9 (Scheme-stable affine translation of the odd selector output)**.** *Let S and S' be cohomologically admissibly comparable finite-cutoff schemes on the same chosen monotone sector. Assume in addition that:*

1. *the canonical anomaly representatives $\mathcal{A}^{(1)}_{k,S}(u)$ and $\mathcal{A}^{(1)}_{k,S'}(u)$ are swap-even;*

2. *there exist chosen* Γ*-stable selector classes*

$$\mathcal{C}^{S}_{\mathrm{sel},\Lambda_k}(u), \qquad \mathcal{C}^{S'}_{\mathrm{sel},\Lambda_k}(u),$$

and chosen selector representatives

$$\widehat{\mathcal{A}}^{(1)}_{k,S}(u) \in \mathcal{C}^{S}_{\mathrm{sel},\Lambda_k}(u), \qquad \widehat{\mathcal{A}}^{(1)}_{k,S'}(u) \in \mathcal{C}^{S'}_{\mathrm{sel},\Lambda_k}(u),$$

and an even sufficiently local cochain $Y_k(u)$ *such that*

$$\widehat{\mathcal{A}}^{(1)}_{k,S'}(u) - \widehat{\mathcal{A}}^{(1)}_{k,S}(u) = \mathcal{A}^{(1)}_{k,S'}(u) - \mathcal{A}^{(1)}_{k,S}(u) + Q_t Y_k(u). \tag{14}$$

Define the odd comparison primitive

$$Y_k^{-}(u) := \frac{1}{2}\Big(Y_k(u) - \Gamma Y_k(u)\Big).$$

Then

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k,S'}\big(u;\widehat{\mathcal{A}}^{(1)}_{k,S'}\big) = \mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k,S}\big(u;\widehat{\mathcal{A}}^{(1)}_{k,S}\big) + Y_k^{-}(u).$$

If $Y_k'(u)$ *is another comparison primitive satisfying* (14) *for the same chosen pair of representatives, then*

$$Y_k'^{-}(u) - Y_k^{-}(u) \in \mathcal{K}^{-}_{\Lambda_k}.$$

Proof. Taking swap-odd parts of (14) and using the swap-evenness of the canonical anomaly representatives gives

$$\widehat{\mathcal{A}}^{(1)}_{k,S',-}(u) - \widehat{\mathcal{A}}^{(1)}_{k,S,-}(u) = Q_t Y_k^{-}(u).$$

The same affine-fiber argument as in proposition 2.8 therefore gives the translation statement by $Y_k^{-}(u)$, and the same odd-kernel argument gives the ambiguity claim for alternative primitives. □

Remark 2.10 (Selector output handed to the downstream notes). The raw selector datum is the affine class

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$$

attached to the chosen selector representative. Its ambiguity descent, transported dominant-response value, and weak-interface image are treated in theorems 3.20, 4.16 and 4.56.

Remark 2.11 (Relation to the Hodge-normalized layer). The Hodge-normalized comparison layer for admissible alternative representatives is treated in theorem 6.10. The present section uses only raw sufficiently local selector classes.

3 Polarized Characterization Interface

This section records the finite-cutoff characterization data extracted from the split/leak construction and used in the 2T Main assumption package Definition 4.2. In this note, an "accepted tick" denotes the local accepted-branch slice at cutoff Λ_k.

For each accepted tick k, write

$$a_{\Lambda_k} := a_k, \qquad S_{\Lambda_k} := a^2_{\Lambda_k}.$$

The variable a_{Λ_k} is the distinguished odd split coordinate identified in sections 1.1 and 1.3; the even quantity S_{Λ_k} is the positive stock that enters the leak law and the split threshold.

Definition 3.1 (Characterization algebra, operational triviality, and even null ideal)**.** Fix an accepted tick k and let $E^-_{\Lambda_k}$ denote the physically odd sufficiently local generator space at cutoff Λ_k, i.e. the odd sector for the transported physical involution Γ. Let $E^+_{\Lambda_k}$ denote the corresponding physically even sufficiently local observable space. Define the core characterization algebra by

$$\mathcal{O}^{\mathrm{char}}_{\Lambda_k} := \mathrm{alg}\{a_{\Lambda_k},\, S_{\Lambda_k}\}. \tag{15}$$

Define the operationally interface-trivial odd subspace by

$$\mathcal{T}^{\mathrm{op}}_{\Lambda_k} := \overline{\Big\{T \in E^-_{\Lambda_k} : \{T, O\}_{\mathrm{P},\Lambda_k} = 0\ \forall O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}\Big\}}. \tag{16}$$

Define the even interface-null ideal by

$$\mathcal{N}^{\mathrm{even}}_{\Lambda_k} := \overline{\Big\{E \in E^+_{\Lambda_k} : \{E, O\}_{\mathrm{P},\Lambda_k} = 0\ \forall O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}\Big\}}. \tag{17}$$

Remark 3.2 (Do not identify the characterization observable with the generator)*.* The odd observable a_{Λ_k} characterizes the active completion-sector channel, but it is not itself the odd Hamiltonian generator class appearing in item **(M4)** of Definition 4.2. That generator is denoted $J^{\mathrm{act}}_{\Lambda_k} \in E^-_{\Lambda_k}$ below. The quotient statement belongs to $[J^{\mathrm{act}}_{\Lambda_k}]$, not to $[a_{\Lambda_k}]$.

Definition 3.3 (Intrinsic covariance-tangent response map)**.** Fix an accepted tick k. Let $\mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ denote the finite-cutoff receiver covariance-tangent space at the UOM reference Gaussian state ρ_{*,Λ_k}, equipped with the Fisher pairing

$$\langle\!\langle \delta C, \delta C' \rangle\!\rangle_{C_{*,\Lambda_k}} := \tfrac{1}{2}\,\mathrm{tr}\big(C^{-1}_{*,\Lambda_k}\delta C\, C^{-1}_{*,\Lambda_k}\delta C'\big).$$

For admissible band-limited receiver probes f, g, let

$$\mathcal{Q}^{\Lambda_k}_{f,g} \in E^+_{\Lambda_k}$$

denote the corresponding sufficiently local quadratic observable, normalized so that on centered Gaussian states with covariance C,

$$\big\langle \mathcal{Q}^{\Lambda_k}_{f,g} \big\rangle_{\rho(C)} = \langle f, Cg\rangle.$$

For $F \in E^-_{\Lambda_k}$, define $\mathcal{R}_{\Lambda_k}(F) \in \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ by the matrix elements

$$\big\langle f, \mathcal{R}_{\Lambda_k}(F) g\big\rangle := \Big\langle \{F, \mathcal{Q}^{\Lambda_k}_{f,g}\}_{\mathrm{P},\Lambda_k}\Big\rangle_{\rho_{*,\Lambda_k}}.$$

We call

$$\mathcal{R}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$$

the *intrinsic covariance-tangent response map.*

Proposition 3.4 (Intrinsic definition and completion independence)**.** *The map $\mathcal{R}_{\Lambda_k}$ of definition 3.3 is continuous and linear. It depends only on the finite-cutoff UOM Peierls package and the UOM covariance/Fisher package. If a Gaussian completion realization is chosen, then $\mathcal{R}_{\Lambda_k}(F)$ is represented by the same completion-sector covariance tangent that one obtains by linearizing the completion quadratures. Thus Gaussian completion is a realization of the response map, not extra structure required to define it.*

Proof. For fixed $F \in E^-_{\Lambda_k}$, the finite-cutoff UOM Peierls package makes

$$\delta_F := \{F, -\}_{\mathrm{P},\Lambda_k}$$

a continuous derivation on the sufficiently local algebra. Applying δ_F to the quadratic probes $\mathcal{Q}^{\Lambda_k}_{f,g}$ and evaluating in the reference Gaussian state produces a continuous symmetric bilinear form in f, g. In the UOM Gaussian radiative sector, such bilinear forms are exactly covariance tangents at C_{*,Λ_k}, equipped with the Fisher metric. Linearity and continuity in F are inherited directly from the Peierls bracket. No completion coordinates are used in this construction.

If a Gaussian completion is later chosen, the same second moments are encoded by the covariance matrix of the completion quadratures. The bilinear form defined above and the completion-sector covariance tangent therefore represent the same infinitesimal response object in two different coordinate systems. □

Definition 3.5 (Physical intrinsic covariance-response map)**.** Fix an accepted tick k. Let

$$\mathcal{T}^{\mathrm{phys}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k} \mathcal{T}^{\mathrm{cov}}_{\Lambda_k} \subset \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}.$$

Define the physical intrinsic covariance-response map by

$$R^{\mathrm{phys}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}.$$

Proposition 3.6 (Physical intrinsic response map)**.** *The map $R^{\mathrm{phys}}_{\Lambda_k}$ of definition 3.5 is continuous and linear.*

Proof. This is immediate from proposition 3.4 and the continuity and linearity of the finite-cutoff physical projector $\Pi_{\mathrm{phys},\Lambda_k}$. □

Proposition 3.7 (Reduction of operational triviality to the active odd coordinate)**.** *Fix an accepted tick k. If $F \in E^-_{\Lambda_k}$ satisfies*

$$\{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0,$$

then $F \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. Since $S_{\Lambda_k} = a^2_{\Lambda_k}$ and the Peierls bracket is a derivation,

$$\{F, S_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = \{F, a^2_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 2\, a_{\Lambda_k} \{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0.$$

The algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k} = \mathrm{alg}\{a_{\Lambda_k}, S_{\Lambda_k}\}$ is generated by a_{Λ_k} and S_{Λ_k}, and the Peierls bracket acts on it by the Leibniz rule. Hence $\{F, O\}_{\mathrm{P},\Lambda_k} = 0$ for every $O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}$, proving $F \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Assumption 3.1 (Active odd generator and intrinsic-response factorization)**.** For each accepted tick k, there exist a distinguished active odd generator

$$J^{\mathrm{act}}_{\Lambda_k} \in E^-_{\Lambda_k}$$

and a nonzero physical covariance tangent

$$C^{\mathrm{act}}_{\Lambda_k} \in \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$$

and a continuous nonzero linear functional

$$\rho_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

normalized by

$$\rho_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1,$$

such that

$$R^{\mathrm{phys}}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = C^{\mathrm{act}}_{\Lambda_k} \neq 0$$

and

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = \rho_{\Lambda_k}(F)\, C^{\mathrm{act}}_{\Lambda_k} \qquad \forall F \in E^-_{\Lambda_k} \tag{18}$$

holds on the strong sector under consideration.

Remark 3.8 (Strong normalization). If one further chooses the Fisher-unit normalization $\widehat{C}^{\mathrm{act}}_{\Lambda_k} := \|C^{\mathrm{act}}_{\Lambda_k}\|^{-1}_{C_{*,\Lambda_k}}\, C^{\mathrm{act}}_{\Lambda_k}$, then (18) sharpens to

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = \widehat{\rho}_{\Lambda_k}(F)\, \widehat{C}^{\mathrm{act}}_{\Lambda_k}$$

with $\widehat{\rho}_{\Lambda_k} := \|C^{\mathrm{act}}_{\Lambda_k}\|_{C_{*,\Lambda_k}}\, \rho_{\Lambda_k}$. The export below does not require this extra normalization; it only needs the scalar response coefficient ρ_{Λ_k} relative to a fixed nonzero active tangent $C^{\mathrm{act}}_{\Lambda_k}$.

Assumption 3.2 (Physical-response sufficiency)**.** For each accepted tick k, the operationally trivial odd directions are exactly the kernel of the physical intrinsic covariance-response map:

$$\ker R^{\mathrm{phys}}_{\Lambda_k} = \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Equivalently, an odd generator has zero physical intrinsic covariance response if and only if it is operationally trivial on the full core characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$.

Remark 3.9 (Why the strong package must be stated intrinsically). The earlier quotient-valued formulation of the strong package in $E^+_{\Lambda_k}/\mathcal{N}^{\mathrm{even}}_{\Lambda_k}$ is not stable. Because the Peierls bracket here is the ordinary derivation bracket and $S_{\Lambda_k} = a^2_{\Lambda_k}$, the even observable $\{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k}$ already Poisson-commutes with both a_{Λ_k} and S_{Λ_k}, hence lies automatically in $\mathcal{N}^{\mathrm{even}}_{\Lambda_k}$. The correct codomain for factorization and kernel identification is therefore the intrinsic physical covariance-response space $\mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$, not the even observable quotient.

Proposition 3.10 (Operational triviality is the kernel of the active response)**.** *Assume assumptions 3.1 and 3.2. Then*

$$\mathcal{T}^{\mathrm{op}}_{\Lambda_k} = \ker \rho_{\Lambda_k}.$$

Proof. Because $C^{\mathrm{act}}_{\Lambda_k} \neq 0$, the factorization identity (18) implies

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = 0 \qquad \Longleftrightarrow \qquad \rho_{\Lambda_k}(F) = 0$$

for every $F \in E^-_{\Lambda_k}$. Thus $\ker R^{\mathrm{phys}}_{\Lambda_k} = \ker \rho_{\Lambda_k}$. Apply assumption 3.2. □

Corollary 3.11 (Rank-one active odd quotient)**.** *Assume assumptions 3.1 and 3.2. Then*

$$E^-_{\Lambda_k}/\mathcal{T}^{\mathrm{op}}_{\Lambda_k} \cong \mathbb{R},$$

and the quotient is generated by the class of the active odd generator $[J^{\mathrm{act}}_{\Lambda_k}]$.

Proof. By proposition 3.10, the quotient is canonically isomorphic to the image of the nonzero functional ρ_{Λ_k}. Since ρ_{Λ_k} is real-valued and $\rho_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1$, its image is $\mathbb{R}$ and the quotient is generated by $[J^{\mathrm{act}}_{\Lambda_k}]$. □

Proposition 3.12 (BRST/gauge-exact compatibility on the characterization layer)**.** *For each accepted tick k, the odd parts of the BRST/gauge-exact directions*

$$Q_t X + Q_u Y$$

lie in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. The exported characterization observables a_{Λ_k} and S_{Λ_k} are physical finite-cutoff observables on the characterization layer. The BRST/gauge-exact odd directions $Q_t X + Q_u Y$ represent redundancies of that physical layer, so their Peierls action vanishes on $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. By (16), their odd parts therefore lie in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Proposition 3.13 (Boundary-admissible divergence compatibility)**.** *Let K^μ be a sufficiently local current whose renormalized boundary flux vanishes on the core characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. Then the odd part of*

$$\partial_\mu K^\mu$$

lies in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. For the admissible currents in the statement, the renormalized Stokes reduction of $\partial_\mu K^\mu$ produces no boundary term on the characterization layer. Hence its Peierls action on every $O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}$ vanishes, and the odd part belongs to $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ by (16). □

Definition 3.14 (Admissible odd counterterms on the characterization layer)**.** For each accepted tick k, an odd sufficiently local counterterm ambiguity

$$\Delta^-_{\mathrm{ct}} \in E^-_{\Lambda_k}$$

is called *admissible on the characterization layer* if its intrinsic covariance response has no physical projection:

$$\Pi_{\mathrm{phys},\Lambda_k}\, \mathcal{R}_{\Lambda_k}(\Delta^-_{\mathrm{ct}}) = 0.$$

Equivalently, admissible odd counterterms are precisely those local scheme shifts whose response is confined to the RP, Weyl, conformal, or other even-null monitor sectors and does not enter the compressed active odd line.

Proposition 3.15 (Counterterm rigidity on the characterization layer)**.** *Assume that the characterization observable a_{Λ_k} is extracted from the physical covariance response so that*

$$\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F) = 0 \quad \Longrightarrow \quad \{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0 \qquad \forall F \in E^-_{\Lambda_k}.$$

Then every admissible odd counterterm ambiguity in the sense of definition 3.14 acts trivially on the core characterization algebra:

$$\Delta^-_{\mathrm{ct}} \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Proof. If $F = \Delta^-_{\mathrm{ct}}$ is admissible, then this physical projection vanishes by definition. The displayed hypothesis therefore gives

$$\{\Delta^-_{\mathrm{ct}}, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0,$$

and since $S_{\Lambda_k} = a^2_{\Lambda_k}$, the derivation property yields

$$\{\Delta^-_{\mathrm{ct}}, S_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0.$$

By proposition 3.7, the first identity already implies $\Delta^-_{\mathrm{ct}} \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Remark 3.16 (Active stiffness versus full calibration). The split/leak analysis determines the active stiffness $\widetilde{\alpha}(\Lambda_k)$ on the distinguished line generated by $J^{\mathrm{act}}_{\Lambda_k}$, with

$$\widetilde{\alpha}(\Lambda_k) \asymp \alpha(\Lambda_k) \sim \Lambda_k^{d+2}$$

by assumption 1.7. What UOM does *not* determine canonically is a positive bilinear form on the whole odd generator space $E^-_{\Lambda_k}$. That extension is auxiliary interface data, not a UOM-dynamical output.

Assumption 3.3 (Auxiliary coercive core on the operationally trivial sector)**.** For each accepted tick k, choose a continuous symmetric coercive bilinear form

$$\langle \cdot, \cdot \rangle^{\mathrm{aux}}_{\Lambda_k} : \mathcal{T}^{\mathrm{op}}_{\Lambda_k} \times \mathcal{T}^{\mathrm{op}}_{\Lambda_k} \to \mathbb{R}$$

whose induced norm Hilbertizes $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proposition 3.17 (Strong-package calibration extension from active stiffness)**.** *Assume assumptions 3.1 to 3.3. Set*

$$\chi_{\Lambda_k} := \rho_{\Lambda_k}, \qquad \mathcal{T}_{\Lambda_k} := \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Then the formula

$$\mathsf{B}_{\Lambda_k}(F, G) := \widetilde{\alpha}(\Lambda_k)\, \chi_{\Lambda_k}(F)\chi_{\Lambda_k}(G) + \big\langle F - \chi_{\Lambda_k}(F) J^{\mathrm{act}}_{\Lambda_k},\, G - \chi_{\Lambda_k}(G) J^{\mathrm{act}}_{\Lambda_k} \big\rangle^{\mathrm{aux}}_{\Lambda_k} \tag{19}$$

defines a continuous symmetric positive definite bilinear form on $E^-_{\Lambda_k}$ satisfying

$$\mathsf{B}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}, J^{\mathrm{act}}_{\Lambda_k}) = \widetilde{\alpha}(\Lambda_k), \qquad \mathsf{B}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}, T) = 0 \quad \forall T \in \mathcal{T}_{\Lambda_k}.$$

Proof. By proposition 3.10, every $F \in E^-_{\Lambda_k}$ has a unique decomposition

$$F = \chi_{\Lambda_k}(F)\, J^{\mathrm{act}}_{\Lambda_k} + T_F, \qquad T_F \in \mathcal{T}_{\Lambda_k}.$$

Equation (19) is therefore well-defined, continuous, and symmetric. Positivity is immediate from $\widetilde{\alpha}(\Lambda_k) > 0$ together with coercivity of $\langle \cdot, \cdot \rangle^{\mathrm{aux}}_{\Lambda_k}$ on $\mathcal{T}_{\Lambda_k}$. If $\mathsf{B}_{\Lambda_k}(F, F) = 0$, then both $\chi_{\Lambda_k}(F) = 0$ and $\langle T_F, T_F \rangle^{\mathrm{aux}}_{\Lambda_k} = 0$, hence $T_F = 0$ and $F = 0$. The displayed normalization and orthogonality properties follow by substitution into (19). □

Corollary 3.18 (Descendant ambiguity compatibility on the characterization layer)**.** *For each accepted tick k, the odd-projected ambiguity directions coming from*

$$Q_t X + Q_u Y, \qquad \partial_\mu K^\mu, \qquad \Delta^-_{\mathrm{ct}}$$

belong to $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ whenever K^μ is taken from the boundary-admissible class of proposition 3.13 and Δ^-_{ct} is admissible in the sense of definition 3.14.

Proof. Combine propositions 3.12, 3.13 and 3.15. □

Theorem 3.19 (Strong rank-one realization of the characterization layer)**.** *Assume assumptions 1.4, 1.6, 1.7 and 3.1 to 3.3. Then for each accepted tick k, the split/leak construction exports the UOM-side data*

$$\big(a_{\Lambda_k},\, S_{\Lambda_k},\, J^{\mathrm{act}}_{\Lambda_k},\, C^{\mathrm{act}}_{\Lambda_k},\, \widetilde{\alpha}(\Lambda_k),\, \rho_{\Lambda_k}\big), \tag{20}$$

where:

1. $\mathcal{O}^{\mathrm{char}}_{\Lambda_k} = \mathrm{alg}\{a_{\Lambda_k}, S_{\Lambda_k}\}$ *is the minimal completion-sector characterization algebra;*

2. *the operationally trivial subspace satisfies*

$$\mathcal{T}_{\Lambda_k} := \mathcal{T}^{\mathrm{op}}_{\Lambda_k} = \ker R^{\mathrm{phys}}_{\Lambda_k} = \ker \rho_{\Lambda_k};$$

3. *the intrinsic physical response is one-dimensional on the strong sector:*

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = \rho_{\Lambda_k}(F)\, C^{\mathrm{act}}_{\Lambda_k}, \qquad R^{\mathrm{phys}}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = C^{\mathrm{act}}_{\Lambda_k} \neq 0;$$

4. *the active odd quotient is rank one and is generated by* $[J^{\mathrm{act}}_{\Lambda_k}]$*;*

5. *the abstract orientation functional of the transported interface is realized by*

$$\chi_{\Lambda_k} := \rho_{\Lambda_k};$$

6. *any admissible calibration form* B_{Λ_k} *in this strong package may be obtained by extending the active stiffness* $\widetilde{\alpha}(\Lambda_k)$ *with proposition 3.17.*

Proof. Definition 3.1 packages the active completion-sector channel extracted from the split/leak construction: a_{Λ_k} is the distinguished odd split coordinate singled out by sections 1.1 and 1.3, while $S_{\Lambda_k} = a^2_{\Lambda_k}$ is the even stock entering (16) and (19). The active odd generator $J^{\mathrm{act}}_{\Lambda_k}$, active physical covariance tangent $C^{\mathrm{act}}_{\Lambda_k}$, and scalar response coefficient ρ_{Λ_k} are exported by assumption 3.1; proposition 3.10 and corollary 3.11 then identify the abstract interface-trivial subspace and the corresponding rank-one quotient. The split/leak law fixes the active stiffness $\widetilde{\alpha}(\Lambda_k)$, and proposition 3.17 upgrades that one-line datum to an admissible full calibration form once the auxiliary coercive core is chosen. Finally, corollary 3.18 places the odd-projected descendant ambiguities in $\mathcal{T}_{\Lambda_k}$, yielding the stated ambiguity compatibility on the characterization layer. □

Theorem 3.20 (Intrinsic weak transported descendant-class interface)**.** *Assume the hypotheses of theorem 4.56 on a connected accepted-branch segment* I*. Then for every* $\Lambda \in I$*, the sufficiently local odd generator layer carries the weak transported interface data*

$$\big(E^-_\Lambda,\, \mathcal{A}^-_\Lambda,\, [\Xi^{(1)}_\Lambda],\, \chi_\Lambda,\, \varepsilon_\Lambda\big)$$

with

$$\chi_\Lambda := \nu^{-1}_\Lambda \rho^{\mathrm{dom}}_\Lambda,$$

and the following conclusions hold.

1. *there exists a representative*

$$J^{\mathrm{desc}}_\Lambda \in [\Xi^{(1)}_\Lambda]$$

such that

$$\rho^{\mathrm{dom}}_\Lambda\big(J^{\mathrm{desc}}_\Lambda\big) \neq 0;$$

equivalently, the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line;

2. *the orientation functional is realized intrinsically on the sufficiently local generator layer:*

$$\mathcal{A}^-_\Lambda \subset \ker \chi_\Lambda, \qquad \chi_\Lambda(J) = \varepsilon_\Lambda \quad \forall J \in [\Xi^{(1)}_\Lambda];$$

3. *along I, the sign ε_Λ is locally constant.*

In particular, the weak transported interface is realized intrinsically, without appeal to completion coordinates, under the stated hypotheses.

Proof. The functional $\rho_\Lambda^{\mathrm{dom}}$ is defined intrinsically from the sufficiently local generator layer by definition 3.3 together with the dominant-line prescription of definition 4.3. Item (1) is exactly the nonzero-overlap conclusion of item (3) in theorem 4.56, item (2) is exactly item (4) there, and item (3) is item (5). □

Theorem 3.21 (Quadratic-probe realization of the weak activity supplement)**.** *Assume the hypotheses of theorem 4.56 on a connected accepted-branch segment I. For each $\Lambda \in I$, define the even quadratic probe sector by*

$$\mathcal{P}_\Lambda^+ := \operatorname{span}\big\{\mathcal{Q}_{f,g}^\Lambda : f, g \text{ admissible band-limited receiver probes}\big\} \subset E_\Lambda^+.$$

Then, for every $F \in E_\Lambda^-$,

$$\{F, A\}_{\mathrm{P},\Lambda} = 0 \qquad \forall A \in \mathcal{P}_\Lambda^+$$

implies

$$\chi_\Lambda(F) = 0.$$

Equivalently, if $\chi_\Lambda(F) \neq 0$, then there exists a sufficiently local even quadratic probe $A \in \mathcal{P}_\Lambda^+$ such that

$$\{F, A\}_{\mathrm{P},\Lambda} \neq 0.$$

In particular, the weak interface data of theorem 3.20 carry a canonical activity witness sector built from the intrinsic quadratic receiver probes.

Proof. Assume

$$\{F, A\}_{\mathrm{P},\Lambda} = 0 \qquad \forall A \in \mathcal{P}_\Lambda^+.$$

In particular,

$$\{F, \mathcal{Q}_{f,g}^\Lambda\}_{\mathrm{P},\Lambda} = 0$$

for every admissible pair f, g. By definition 3.3, this means

$$\big\langle f, \mathcal{R}_\Lambda(F) g\big\rangle = \Big\langle \{F, \mathcal{Q}_{f,g}^\Lambda\}_{\mathrm{P},\Lambda} \Big\rangle_{\rho_{*,\Lambda}} = 0$$

for all admissible f, g, hence $\mathcal{R}_\Lambda(F) = 0$. Therefore

$$\Pi_{\mathrm{phys},\Lambda} \mathcal{R}_\Lambda(F) = 0,$$

and the definition of the raw dominant-response coefficient in definition 4.3 gives

$$\widetilde{\rho}_\Lambda^{\mathrm{dom}}(F) = 0.$$

By item (1) of theorem 4.56, $\delta C_\Lambda^{\mathrm{LoG}} \neq 0$, so dominant-line realization holds and $\rho_\Lambda^{\mathrm{dom}}$ is well defined. Hence

$$\rho_\Lambda^{\mathrm{dom}}(F) = 0.$$

Now apply theorem 3.20, which identifies $\chi_\Lambda = \nu_\Lambda^{-1} \rho_\Lambda^{\mathrm{dom}}$, to obtain

$$\chi_\Lambda(F) = 0.$$

The contrapositive gives the equivalent detection statement. □

Remark 3.22 (Correct scalar form of representative comparisons). The weak export theorem should be read ρ^{dom}-first on the ambiguity quotient. Because

$$\chi_\Lambda := \nu_\Lambda^{-1}\rho_\Lambda^{\mathrm{dom}},$$

any comparison statement that depends only on dominant pairing is naturally a statement about the descended scalar

$$\overline{\rho}_\Lambda^{\mathrm{dom}} : E_\Lambda^- / \mathcal{A}_\Lambda^- \to \mathbb{R}$$

or, equivalently, about its value on the transported descendant affine class. Such a statement may be rewritten in χ_Λ form only when the normalization is made explicit; for example, if $\nu_\Lambda = 1$, then the same scalar value is represented by a descended $\bar{\chi}_\Lambda$. Without that normalization the stronger unscaled χ-compatibility form is false in general.

Corollary 3.23 (UOM-stiffness specialization of the weak transported calibration)**.** *Assume assumption 1.7 and fix an accepted tick k for which the conclusions of theorem 3.20 hold. Suppose* $\ker \chi_{\Lambda_k}$ *admits a continuous symmetric coercive bilinear form whose induced norm Hilbertizes it, and choose any*

$$J^\sharp_{\Lambda_k} \in E^-_{\Lambda_k} \qquad \textit{with} \qquad \chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1.$$

Then proposition 4.58 may be applied with

$$\beta_{\Lambda_k} := \widetilde{\alpha}(\Lambda_k)$$

to produce a continuous symmetric positive definite bilinear form

$$\mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k} : E^-_{\Lambda_k} \times E^-_{\Lambda_k} \to \mathbb{R}$$

such that

$$\mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k}(J^\sharp_{\Lambda_k}, J^\sharp_{\Lambda_k}) = \widetilde{\alpha}(\Lambda_k), \qquad \mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0 \quad \forall K \in \ker \chi_{\Lambda_k}.$$

In particular,

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \chi_{\Lambda_k}$$

implies

$$\mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k}(J^\sharp_{\Lambda_k}, A) = 0 \qquad \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Proof. By remark 3.16, the split/leak package fixes the positive stiffness scale $\widetilde{\alpha}(\Lambda_k)$, and assumption 1.7 records its positivity and asymptotic size. Apply proposition 4.58 with $\beta_{\Lambda_k} = \widetilde{\alpha}(\Lambda_k)$. The final orthogonality statement follows from item (2) of theorem 3.20. □

Remark 3.24 (Inherited ledger compatibility). The inherited commutative ledger algebra is intentionally not part of the core characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. Its role is separate: the split model requires the active odd representative to be completion-sector nontrivial while acting trivially, to leading order, on the inherited classical ledger. This is the UOM-side compatibility statement behind the slogan "oddness in the completion sector, classicality in the accessible ledger."

4 Finite-Cutoff Active Response Transport Theorem

This section assembles a finite-cutoff transport theorem for the odd descendant / Hamiltonization lane of the 2T Main assumption package Definition 4.2. Accordingly, the phrase "accepted tick" remains literal UOM cutoff language throughout this note: the transport theorem is pointwise on the accepted branch at fixed Λ_k.

Remark 4.1 (Physical odd sector and branchwise ambient-space conventions). Throughout this note, the superscript $^-$ refers to physical Γ-odd parity on the characterization layer, not to cohomological oddness. Every ambiguity direction appearing below is therefore understood after passage to the physically odd characterization sector $E^-_{\Lambda_k}$.

All branchwise continuity statements are made after the standard accepted-branch identifications of the relevant finite-cutoff source spaces, covariance-tangent spaces, and physical subspaces with fixed ambient finite-cutoff spaces. Operator-norm and Fisher-norm continuity are always understood with respect to those identifications.

Definition 4.2 (Odd ambiguity subspace on the characterization layer)**.** For each accepted tick k, let

$$\mathcal{A}^-_{\Lambda_k} \subset E^-_{\Lambda_k}$$

denote the closed linear span of the odd characterization-layer projections of the descendant ambiguity directions

$$\big(Q_t X + Q_u Y\big)^-, \qquad \big(\partial_\mu K^\mu\big)^-, \qquad \Delta^-_{\mathrm{ct}},$$

where K^μ is boundary-admissible on the characterization layer and Δ^-_{ct} is admissible in the sense of definition 3.14. Equivalently, $\mathcal{A}^-_{\Lambda_k}$ is the odd ambiguity space already placed inside $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ by corollary 3.18.

Definition 4.3 (Dominant compressed response line and scalar coefficient)**.** Fix an accepted tick k. Let

$$u^{\mathrm{LoG}}_{\Lambda_k}$$

and

$$P^{\mathrm{LoG}}_{-,\Lambda_k}$$

be the canonically selected compressed odd line and projector furnished by corollary 1.10. Define the physical covariance tangent induced by that line by

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k}\, \mathsf{T}_{\Lambda_k}[u^{\mathrm{LoG}}_{\Lambda_k}],$$

and, whenever this tangent is nonzero, write

$$L^{\mathrm{dom}}_{\Lambda_k} := \mathbb{R}\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$$

for its Fisher-normalized line. The raw dominant-response coefficient is

$$\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}(F) := \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}, \qquad F \in E^-_{\Lambda_k}. \tag{21}$$

Whenever $\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0$, equivalently whenever dominant-line realization holds at Λ_k, define the normalized dominant-response coefficient by

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F) := \kappa^{-1/2}_{\Lambda_k}\, \widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}(F), \qquad F \in E^-_{\Lambda_k}. \tag{22}$$

Proposition 4.4 (Well-posedness and continuity of the dominant-response coefficients)**.** *Fix an accepted tick k. Then the raw dominant-response coefficient*

$$\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

is continuous and linear. If dominant-line realization holds at Λ_k, then the normalized dominant-response coefficient

$$\rho^{\mathrm{dom}}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

is also continuous and linear.

Proof. By proposition 3.4, the intrinsic response map $\mathcal{R}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ is continuous and linear. Pairing its physical projection against the fixed covariance tangent $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ in the Fisher metric therefore gives a continuous linear functional, which is exactly $\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$. If dominant-line realization holds, then $\kappa_{\Lambda_k} > 0$ by proposition 4.7, so $\rho^{\mathrm{dom}}_{\Lambda_k}$ is obtained from $\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$ by multiplication by the fixed scalar $\kappa^{-1/2}_{\Lambda_k}$. □

Dominant-line realization

The compressed spectral-selection note already supplies the one-dimensional source line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. The present transport note needs only the nonvanishing and branchwise continuity of its *physical* covariance push-forward.

Definition 4.5 (Compressed-line transversality)**.** For an accepted tick k, say that the welded response is *transversely physical on the compressed line* if

$$\Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k} \restriction_{\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}} \neq 0. \tag{23}$$

Equivalently, the unique generator $u^{\mathrm{LoG}}_{\Lambda_k}$ of the compressed odd line has nonzero physical covariance response.

Definition 4.6 (Dominant-response norm)**.** Whenever $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ is defined, set

$$\kappa_{\Lambda_k} := \left\langle\!\left\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \right\rangle\!\right\rangle_{C_{*,\Lambda_k}}. \tag{24}$$

This is the Fisher squared norm of the physical covariance tangent induced by the compressed welded LoG source line.

Proposition 4.7 (Equivalent formulations of dominant-line realization)**.** *For a fixed accepted tick k, the following are equivalent.*

1. *the compressed welded LoG line is transversely physical in the sense of definition 4.5;*
2. *the distinguished physical covariance tangent is nonzero,*

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0;$$

3. *the dominant-response norm is strictly positive,*

$$\kappa_{\Lambda_k} > 0;$$

4. *there exists a physical covariance tangent* $B_{\Lambda_k} \in \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$ *such that*

$$\langle\!\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, B_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}} \neq 0;$$

5. *in any Gaussian completion realization of corollary 1.11, the completion-sector covariance tangent representing* $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ *is nonzero.*

Proof. The equivalence of (1) and (2) is immediate from (23) and the definition

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} = \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[u^{\mathrm{LoG}}_{\Lambda_k}],$$

because $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$ is the one-dimensional line generated by $u^{\mathrm{LoG}}_{\Lambda_k}$.

The equivalence of (2) and (3) follows from positive definiteness of the Fisher metric on the covariance tangent space: a vector has positive Fisher norm if and only if it is nonzero.

Statement (3) implies (4) by taking $B_{\Lambda_k} := \delta C^{\mathrm{LoG}}_{\Lambda_k}$, which belongs to the physical tangent space by construction. Conversely, if (4) holds then $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ cannot vanish, so (2) holds.

Finally, proposition 3.4 and corollary 1.11 identify the abstract covariance tangent with any Gaussian completion representative of the same tangent object. Therefore the abstract tangent is nonzero exactly when its completion realization is nonzero, giving the equivalence of (2) and (5). □

Theorem 4.8 (Branchwise realization and persistence of the dominant line)**.** *Let* I *be a connected accepted-branch segment in cutoff space. Assume that:*

1. *the compressed projector* $P^{\mathrm{LoG}}_{-,\Lambda}$ *varies continuously on* I*;*
2. *the physical welded response operator*

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_{\Lambda}$$

is continuous on I *in the operator topology induced by the Fisher norm on covariance tangents;*

3. *there exists a constant* $c_I > 0$ *such that*

$$\kappa_\Lambda \geq c_I \qquad \forall \Lambda \in I.$$

Then:

1. $\delta C^{\mathrm{LoG}}_{\Lambda} \neq 0$ *for every* $\Lambda \in I$*;*
2. *the normalized dominant vector*

$$\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda} := \kappa_\Lambda^{-1/2}\, \delta C^{\mathrm{LoG}}_{\Lambda}$$

is continuous on I*;*

3. *the dominant physical line*

$$L^{\mathrm{dom}}_{\Lambda} = \mathbb{R}\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda}$$

is canonically transported along I*;*

4. *the Fisher-orthogonal rank-one projector*

$$\Pi^{\mathrm{dom}}_{\Lambda} := \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda} \otimes \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda}$$

depends continuously on $\Lambda \in I$*.*

Proof. By corollary 1.10, the compressed projector $P^{\mathrm{LoG}}_{-,\Lambda}$ and hence the normalized source vector $u^{\mathrm{LoG}}_{\Lambda}$ vary continuously on the branch. By the assumed continuity of $\Pi_{\mathrm{phys},\Lambda}\mathsf{T}_{\Lambda}$, the map

$$\Lambda \longmapsto \delta C^{\mathrm{LoG}}_{\Lambda} = \Pi_{\mathrm{phys},\Lambda}\mathsf{T}_{\Lambda}[u^{\mathrm{LoG}}_{\Lambda}]$$

is continuous in the Fisher norm.

The lower bound $\kappa_{\Lambda} \geq c_I > 0$ implies $\delta C^{\mathrm{LoG}}_{\Lambda} \neq 0$ for every $\Lambda \in I$, proving item (1). Since the normalization map $v \mapsto v/\|v\|_{C_*}$ is continuous on the complement of the zero section, continuity of $\delta C^{\mathrm{LoG}}_{\Lambda}$ and the positive lower bound on its norm imply continuity of $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda}$, proving item (2). Items (3) and (4) are immediate consequences of item (2). □

Corollary 4.9 (Local persistence from one-tick transversality)**.** *Assume the continuity hypotheses of theorem 4.8. If the compressed line is transversely physical at one accepted tick Λ_{k_0}, then there exists a neighborhood I_{k_0} of Λ_{k_0} on the same accepted branch such that*

$$\delta C^{\mathrm{LoG}}_{\Lambda} \neq 0 \qquad \forall \Lambda \in I_{k_0}.$$

Hence the dominant physical line is locally well defined and continuously transported near Λ_{k_0}.

Proof. By proposition 4.7, transversality at Λ_{k_0} is equivalent to $\kappa_{\Lambda_{k_0}} > 0$. Continuity of κ_{Λ} then gives a neighborhood I_{k_0} and a constant $c_{k_0} > 0$ such that $\kappa_{\Lambda} \geq c_{k_0}$ for all $\Lambda \in I_{k_0}$. Apply theorem 4.8. □

Remark 4.10 (Practical witness criteria)*.* For applications, any one of the following suffices to discharge dominant-line realization at a fixed accepted tick:

1. a direct lower bound $\kappa_{\Lambda_k} > 0$;
2. a physical witness tangent $B_{\Lambda_k} \in \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$ with nonzero Fisher pairing against $\delta C^{\mathrm{LoG}}_{\Lambda_k}$;
3. a Gaussian completion calculation showing that the completion covariance of the realized active odd mode changes nontrivially under the compressed welded LoG source.

Each witness proves the same theorem-level fact by proposition 4.7.

Descent to the ambiguity quotient

Once the dominant line has been realized, the next task is purely structural: show that the dominant-response coefficient is insensitive to ambiguity moves. This is the exact point at which the transported descendant class becomes an affine class in the sense required by Main rather than a collection of representatives.

Definition 4.11 (Dominant-response quotient and descended descendant point)**.** For each accepted tick k, define the dominant-response quotient by

$$\mathfrak{D}^{-}_{\Lambda_k} := E^{-}_{\Lambda_k}/\mathcal{A}^{-}_{\Lambda_k},$$

with quotient map

$$q_{\Lambda_k} : E^{-}_{\Lambda_k} \to \mathfrak{D}^{-}_{\Lambda_k}.$$

If

$$[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^{-}_{\Lambda_k},$$

then its image in the quotient is a single point,

$$[\![\Xi^{(1)}_{\Lambda_k}]\!] := q_{\Lambda_k}\big(J_{0,\Lambda_k}\big) \in \mathfrak{D}^-_{\Lambda_k},$$

independent of the chosen representative J_{0,Λ_k}.

Proposition 4.12 (Universal descent criterion for the dominant-response coefficient)**.** *Assume dominant-line realization at an accepted tick k, so that $\rho^{\mathrm{dom}}_{\Lambda_k}$ is defined by (22). Then the following are equivalent.*

1. *there exists a unique continuous linear functional*

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathbb{R}$$

such that

$$\rho^{\mathrm{dom}}_{\Lambda_k} = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \circ q_{\Lambda_k};$$

2. *the ambiguity subspace is contained in the kernel:*

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k};$$

3. *the dominant-response coefficient is constant on each ambiguity coset:*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F + A) = \rho^{\mathrm{dom}}_{\Lambda_k}(F) \qquad \forall F \in E^-_{\Lambda_k},\ \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Proof. The equivalence of (2) and (3) is immediate from linearity:

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F + A) - \rho^{\mathrm{dom}}_{\Lambda_k}(F) = \rho^{\mathrm{dom}}_{\Lambda_k}(A).$$

If (2) holds, define

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(q_{\Lambda_k}(F)\big) := \rho^{\mathrm{dom}}_{\Lambda_k}(F).$$

This is well defined precisely because changing representatives by an element of $\mathcal{A}^-_{\Lambda_k}$ does not change the value of $\rho^{\mathrm{dom}}_{\Lambda_k}$. Linearity and continuity are inherited from $\rho^{\mathrm{dom}}_{\Lambda_k}$, and uniqueness follows because q_{Λ_k} is surjective.

Conversely, if (1) holds and $A \in \mathcal{A}^-_{\Lambda_k}$, then $q_{\Lambda_k}(A) = 0$, hence

$$\rho^{\mathrm{dom}}_{\Lambda_k}(A) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(q_{\Lambda_k}(A)\big) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(0) = 0.$$

Thus (1) implies (2). □

Corollary 4.13 (Constancy on the transported descendant affine class)**.** *Assume proposition 4.12. Then*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \qquad \forall J \in [\Xi^{(1)}_{\Lambda_k}].$$

In particular, once the descended quotient functional exists, the descendant affine class carries a single unambiguous dominant-response value.

Proof. Every $J \in [\Xi^{(1)}_{\Lambda_k}]$ has the form $J = J_{0,\Lambda_k} + A$ with $A \in \mathcal{A}^-_{\Lambda_k}$, so

$$q_{\Lambda_k}(J) = q_{\Lambda_k}(J_{0,\Lambda_k}) = [\![\Xi^{(1)}_{\Lambda_k}]\!].$$

Apply the factorization identity of proposition 4.12. □

Remark 4.14 (False stronger form and normalized special case). The quotient statement above is intrinsically a statement about the descended dominant scalar

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}([\![\Xi^{(1)}_{\Lambda_k}]\!]),$$

not about a bare χ-compatibility condition. At the weak transported-interface layer one has

$$\chi_{\Lambda_k} = \nu^{-1}_{\Lambda_k}\rho^{\mathrm{dom}}_{\Lambda_k},$$

so a representative-search condition of the form

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = s$$

is equivalent to

$$\chi_{\Lambda_k}(J) = \nu^{-1}_{\Lambda_k}s,$$

not in general to $\chi_{\Lambda_k}(J) = s$. Only after an explicit normalization, for example $\nu_{\Lambda_k} = 1$, may the same scalar comparison be rewritten in descended $\bar{\chi}_{\Lambda_k}$ form without changing the value. The finite-cutoff Wolfram audit contains exact countermodels to the stronger unscaled χ-level claim.

Proposition 4.15 (Generator-by-generator reduction)**.** *Assume dominant-line realization at an accepted tick k. Suppose the dominant-response coefficient vanishes on the three generating ambiguity families:*

1. *for every odd-projected BRST/gauge-exact direction $\big(Q_tX + Q_uY\big)^-$,*

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(\big(Q_tX + Q_uY\big)^-\big) = 0;$$

2. *for every odd-projected boundary-admissible divergence $\big(\partial_\mu K^\mu\big)^-$,*

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(\big(\partial_\mu K^\mu\big)^-\big) = 0;$$

3. *for every admissible odd counterterm Δ^-_{ct},*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(\Delta^-_{\mathrm{ct}}) = 0.$$

Then

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k},$$

hence $\rho^{\mathrm{dom}}_{\Lambda_k}$ descends uniquely to the quotient $\mathfrak{D}^-_{\Lambda_k}$.

Proof. By definition 4.2, $\mathcal{A}^-_{\Lambda_k}$ is the closed linear span of the three families listed in the statement. Since $\rho^{\mathrm{dom}}_{\Lambda_k}$ is linear, it vanishes on the algebraic span of those generators. Since it is also continuous, it vanishes on the closure of that span. Therefore every element of $\mathcal{A}^-_{\Lambda_k}$ lies in the kernel. Now apply proposition 4.12. □

Theorem 4.16 (Descent to the ambiguity quotient)**.** *Fix an accepted tick k and assume dominant-line realization. If the three vanishing statements of proposition 4.15 hold, then there exists a unique descended dominant functional*

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathbb{R}$$

such that

$$\rho^{\mathrm{dom}}_{\Lambda_k} = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \circ q_{\Lambda_k},$$

and the transported descendant affine class has a single well-defined quotient value

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}([\![\Xi^{(1)}_{\Lambda_k}]\!]).$$

Proof. Apply proposition 4.15 and corollary 4.13. □

Nonzero overlap on the descendant class

After dominant-line realization and quotient descent, the descendant class defines a single point in the ambiguity quotient, and the dominant-response functional descends to that quotient. This subsection studies whether that descended descendant point is *active*, i.e. not annihilated by the descended dominant functional.

Definition 4.17 (Dominant covariance projector)**.** Assume dominant-line realization at an accepted tick k. Define the Fisher-orthogonal rank-one projector onto the dominant physical line by

$$\Pi^{\mathrm{dom}}_{\Lambda_k}(X) := \langle\!\langle X,\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}} \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}, \qquad X \in \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}. \tag{25}$$

Proposition 4.18 (Representative-level witness criteria)**.** *Assume dominant-line realization at an accepted tick k. For any odd generator $F \in E^-_{\Lambda_k}$, the following are equivalent.*

1. *the dominant-response coefficient is nonzero:*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F) \neq 0;$$

2. *the dominant projection of the physical covariance response is nonzero:*

$$\Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F) \neq 0;$$

3. *the physical covariance response has nonzero Fisher pairing with the dominant tangent:*

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}} \neq 0;$$

4. *in any Gaussian completion realization of corollary 1.11, the completion representative of $\Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F)$ has nonzero component on the realized active odd line.*

Proof. By definition 4.17,

$$\Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F) = \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}.$$

Since $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \neq 0$, this proves the equivalence of (1) and (2).

Statement (3) is equivalent to (1) because

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}} = \|\delta C^{\mathrm{LoG}}_{\Lambda_k}\|_{C_{*,\Lambda_k}}\, \rho^{\mathrm{dom}}_{\Lambda_k}(F),$$

and the prefactor is strictly positive by dominant-line realization.

Finally, by proposition 3.4 and corollary 1.11, the abstract covariance tangent and its Gaussian completion representative describe the same physical response object in different coordinates. Therefore the abstract dominant projection is nonzero if and only if the completion representative has nonzero component on the realized active line, proving the equivalence of (2) and (4). □

Definition 4.19 (Descendant overlap margin)**.** Assume dominant-line realization and quotient descent at an accepted tick k. Define the descendant overlap margin by

$$\nu_{\Lambda_k} := \left| \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \right|. \tag{26}$$

We say that the transported descendant class is *dominant-active* if $\nu_{\Lambda_k} > 0$.

Proposition 4.20 (Equivalent formulations of nonzero overlap on the descendant class)**.** *Assume dominant-line realization and quotient descent at an accepted tick k. Then the following are equivalent.*

1. *the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line;*
2. *there exists one representative $J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ such that*
$$\Pi^{\mathrm{dom}}_{\Lambda_k}\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(J^{\mathrm{desc}}_{\Lambda_k}) \neq 0;$$
3. *every representative $J \in [\Xi^{(1)}_{\Lambda_k}]$ has the same nonzero dominant-response value:*
$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \neq 0;$$
4. *the descended descendant point is not annihilated by the descended dominant functional:*
$$[\![\Xi^{(1)}_{\Lambda_k}]\!] \notin \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k};$$
5. *the descendant overlap margin is strictly positive:*
$$\nu_{\Lambda_k} > 0.$$

Proof. The equivalence of (1) and (2) follows immediately from proposition 4.18.

If (2) holds, then $\rho^{\mathrm{dom}}_{\Lambda_k}(J^{\mathrm{desc}}_{\Lambda_k}) \neq 0$ by proposition 4.18. By corollary 4.13, every other representative of $[\Xi^{(1)}_{\Lambda_k}]$ has the same value, proving (3). Conversely, (3) obviously implies (1).

The equivalence of (3) and (4) is immediate from the identity
$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \qquad \forall J \in [\Xi^{(1)}_{\Lambda_k}]$$
of corollary 4.13.

Finally, (4) and (5) are equivalent by the definition of ν_{Λ_k} in definition 4.19. □

Theorem 4.21 (Nonzero overlap on the descendant class)**.** *Fix an accepted tick k, and assume dominant-line realization together with quotient descent. If any one of the equivalent conditions in proposition 4.20 holds, then the following conclusions obtain. In particular, it is enough to construct a single representative*
$$J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$
whose dominant projected covariance response is nonzero.

When this holds, the quotient descendant point is dominant-active, the scalar value
$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)$$
is nonzero, and the sign
$$\varepsilon_{\Lambda_k} := \mathrm{sgn}\Big(\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)\Big) \in \{\pm 1\} \tag{27}$$
is well defined. Thus the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line.

Proof. The equivalence package is exactly proposition 4.20. Once one representative has nonzero dominant projected response, the same proposition gives the nonzero scalar value on the descended descendant point. The sign in (27) is therefore well defined. □

Conditional mod-null reentry on the selected scalar line

Definition 4.22 (Parity-stable mod-null subspace transverse to the selected scalar line)**.** Fix an accepted tick k, assume dominant-line realization and quotient descent, and let

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line in the ambiguity quotient. A *parity-stable mod-null datum transverse to* $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ consists of a linear subspace

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

such that:

(a) $N^{\mathrm{mn}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$;

(b) the selected scalar line is transverse to $N^{\mathrm{mn}}_{\Lambda_k}$:

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}.$$

Define

$$\mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k} := \mathfrak{D}^-_{\Lambda_k}/N^{\mathrm{mn}}_{\Lambda_k}$$

with quotient map

$$\pi^{\mathrm{mn}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k}.$$

Because $\mathfrak{D}^-_{\Lambda_k}$ is itself a quotient of the physical Γ-odd sector, the descended parity acts on $\mathfrak{D}^-_{\Lambda_k}$ by $-\mathrm{id}$. Thus the parity-stability clause yields a descended involution

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\pi^{\mathrm{mn}}_{\Lambda_k}(x)\big) := \pi^{\mathrm{mn}}_{\Lambda_k}(-x), \qquad x \in \mathfrak{D}^-_{\Lambda_k}.$$

Definition 4.23 (Dominant-null parity-stable mod-null datum)**.** Fix an accepted tick k, assume dominant-line realization and quotient descent, and let

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line in the ambiguity quotient. A parity-stable mod-null datum

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

is called *dominant-null* if it is contained in the kernel of the descended dominant functional:

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}.$$

It is understood that $N^{\mathrm{mn}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$, and that the associated quotient $\mathfrak{D}^-_{\Lambda_k}/N^{\mathrm{mn}}_{\Lambda_k}$ carries the descended involution from definition 4.22. The canonical dominant-kernel choice is isolated separately in definition 4.24. The transversality lemma below uses only this kernel-subordination property.

Definition 4.24 (Canonical dominant-null datum)**.** Fix an accepted tick k and assume dominant-line realization together with quotient descent. Define the *canonical dominant-null datum* by

$$N^{\mathrm{dom-null}}_{\Lambda_k} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}.$$

This datum depends only on the descended dominant functional on the ambiguity quotient. No additional reduced-welded or completion-side choice is built into the definition.

Lemma 4.25 (Dominant-kernel criterion for selected-line transversality)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}) \neq 0.$$

If

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

is a dominant-null parity-stable mod-null datum in the sense of definition 4.23, then

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}.$$

So every dominant-null parity-stable mod-null datum is automatically transverse to the selected scalar line.

Proof. Let $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k}$. Because $\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}$, there exists $a \in \mathbb{R}$ with $x = a\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}$. Since $N^{\mathrm{mn}}_{\Lambda_k} \subset \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}$, we have

$$0 = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(x) = a\,\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}).$$

The coefficient $\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})$ is nonzero by hypothesis, so $a = 0$. Therefore $x = 0$, proving $\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$. □

Theorem 4.26 (The dominant kernel is the canonical dominant-null datum)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}) \neq 0.$$

Define

$$N^{\mathrm{dom-null}}_{\Lambda_k} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}.$$

Then:

1. *$N^{\mathrm{dom-null}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$;*
2. *$N^{\mathrm{dom-null}}_{\Lambda_k}$ is a dominant-null parity-stable mod-null datum in the sense of definition 4.23;*
3. *the selected scalar line is automatically transverse to the canonical dominant-null datum:*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{dom-null}}_{\Lambda_k} = \{0\}.$$

Hence the canonical dominant kernel is admissible as the mod-null datum for the conditional reentry theorem.

Proof. Let $x \in N^{\mathrm{dom-null}}_{\Lambda_k}$. Because the descended physical parity acts on $\mathfrak{D}^-_{\Lambda_k}$ by $-\mathrm{id}$, we have

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(-x) = -\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(x) = 0.$$

So $-x \in N^{\mathrm{dom-null}}_{\Lambda_k}$, proving parity stability. Item (2) is then exactly definition 4.23 with $N^{\mathrm{mn}}_{\Lambda_k} = N^{\mathrm{dom-null}}_{\Lambda_k}$. Item (3) is lemma 4.25 applied to this canonical choice. □

Lemma 4.27 (Transversality reduces mod-null reentry to quotient construction)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big) \neq 0.$$

Assume moreover a parity-stable mod-null datum transverse to $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ in the sense of definition 4.22. Define

$$\ell^{\mathrm{sel,mn}}_{\Lambda_k} := \pi^{\mathrm{mn}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big)$$

and

$$L^{\mathrm{sel,mn}}_{\Lambda_k} := \mathbb{R}\,\ell^{\mathrm{sel,mn}}_{\Lambda_k} \subset \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k}.$$

Then:

1. *the restriction of $\pi^{\mathrm{mn}}_{\Lambda_k}$ to $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ is injective;*
2. *$L^{\mathrm{sel,mn}}_{\Lambda_k}$ is a well-defined one-dimensional line in the mod-null quotient, generated by the nonzero class $\ell^{\mathrm{sel,mn}}_{\Lambda_k}$;*
3. *the mod-null selected realization map*

$$\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k} : L^{\mathrm{sel,mn}}_{\Lambda_k} \longrightarrow \mathbb{R}\,u^{\mathrm{LoG}}_{\Lambda_k}, \qquad \mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}\big(a\,\ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) := a\,\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big)\,u^{\mathrm{LoG}}_{\Lambda_k}$$

   is well defined and faithful;
4. *the descended involution acts by sign:*

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) = -\ell^{\mathrm{sel,mn}}_{\Lambda_k}.$$

In particular, once the transversality condition $\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$ is known, the remaining mod-null reentry step reduces to quotient construction plus parity descent.

Proof. If $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k}$ and $\pi^{\mathrm{mn}}_{\Lambda_k}(x) = 0$, then $x \in N^{\mathrm{mn}}_{\Lambda_k}$ by definition of the quotient kernel. Hence $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$, which proves injectivity. Since $\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \neq 0$, its image $\ell^{\mathrm{sel,mn}}_{\Lambda_k}$ is therefore nonzero, and the image of a one-dimensional line under an injective linear map is again one-dimensional. This proves items (1) and (2).

Injectivity on $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ implies that every element of $L^{\mathrm{sel,mn}}_{\Lambda_k}$ has a unique expression $a\,\ell^{\mathrm{sel,mn}}_{\Lambda_k}$, so the displayed formula defines $\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}$ unambiguously. Its coefficient $\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})$ is nonzero by hypothesis, and $u^{\mathrm{LoG}}_{\Lambda_k} \neq 0$ by corollary 1.10. Therefore $\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}(a\,\ell^{\mathrm{sel,mn}}_{\Lambda_k}) = 0$ implies $a = 0$, so the map is faithful. This proves item (3).

By construction of the descended involution,

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}(\ell^{\mathrm{sel,mn}}_{\Lambda_k}) = \Gamma^{\mathrm{mn}}_{\Lambda_k}\Big(\pi^{\mathrm{mn}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})\Big) = \pi^{\mathrm{mn}}_{\Lambda_k}(-\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}) = -\ell^{\mathrm{sel,mn}}_{\Lambda_k},$$

which proves item (4). □

Theorem 4.28 (Conditional mod-null reentry on the selected scalar line)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses and notation of* **??**. *In particular, let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \subset \mathfrak{D}^-_{\Lambda_{k+1}}$$

be the nonzero selected scalar descendant line supplied there. Assume moreover a parity-stable mod-null datum transverse to $\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}}$ in the sense of definition 4.22. Define

$$\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}} := \pi^{\mathrm{mn}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}})$$

and

$$L^{\mathrm{sel,mn}}_{\Lambda_{k+1}} := \mathbb{R}\,\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}} \subset \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_{k+1}}.$$

Then:

1. *$L^{\mathrm{sel,mn}}_{\Lambda_{k+1}}$ is a well-defined one-dimensional selected descendant line in the mod-null quotient;*
2. *the induced mod-null selected realization map*

$$\mathfrak{R}^{\mathrm{mn}}_{\Lambda_{k+1}} : L^{\mathrm{sel,mn}}_{\Lambda_{k+1}} \longrightarrow \mathbb{R}\,u^{\mathrm{LoG}}_{\Lambda_{k+1}}, \qquad \mathfrak{R}^{\mathrm{mn}}_{\Lambda_{k+1}}(a\,\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}) := a\,\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}})\,u^{\mathrm{LoG}}_{\Lambda_{k+1}}$$

   is well defined and faithful;

3. *the descended involution acts by sign:*

$$\Gamma^{\mathrm{mn}}_{\Lambda_{k+1}}(\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}) = -\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}.$$

Proof. By **??**,

$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0 \qquad \text{and} \qquad \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) \neq 0.$$

So every hypothesis of lemma 4.27 is in force at Λ_{k+1}. Applying that lemma gives items (1)–(3) immediately. □

Corollary 4.29 (Canonical dominant-kernel discharge of the mod-null reentry step)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses and notation of* **??**. *Define the canonical dominant-null datum*

$$N^{\mathrm{dom-null}}_{\Lambda_{k+1}} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}} \subset \mathfrak{D}^-_{\Lambda_{k+1}}.$$

Then $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is admissible as the mod-null datum of theorem 4.28, the transversality hypothesis of that theorem is automatic, and hence the conclusions of that theorem hold with this canonical datum.

Proof. By **??**,

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) \neq 0.$$

So theorem 4.26 applies at Λ_{k+1}. It shows that $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is a dominant-null parity-stable mod-null datum and that

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} \cap N^{\mathrm{dom-null}}_{\Lambda_{k+1}} = \{0\}.$$

This is exactly the admissibility and transversality clause required by theorem 4.28, so that theorem now applies directly with the canonical datum. □

Remark 4.30 (Conditional character of the reentry step). theorem 4.28 is a quotient-level reentry statement. After lemma 4.25, theorem 4.26, and corollary 4.29, the canonical dominant-null datum is fixed as $\ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}$ on $\mathfrak{D}^{-}_{\Lambda_{k+1}}$. What remains is passage to the mod-null quotient together with parity descent on the odd line. Accordingly, the theorem should not be read as a statement on the raw descendant ambiguity quotient or on any stronger physical quotient.

Branchwise orientation transport

This subsection studies transport of the descended descendant sign along a connected accepted branch. The sign can be transported once the descended descendant value varies continuously with the cutoff. The compressed spectral-selection theorem already gives continuity of the dominant projector, and theorem 4.8 gives continuity of the dominant physical line. We combine those inputs with continuity of the projected descendant response.

Definition 4.31 (Branchwise descendant overlap function)**.** Let I be a connected accepted-branch segment. Assume dominant-line realization and quotient descent for every $\Lambda \in I$. Suppose there is a branchwise choice of representatives

$$J^{\mathrm{desc}}_{\Lambda} \in [\Xi^{(1)}_{\Lambda}] \qquad (\Lambda \in I)$$

such that the projected covariance response

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda}\mathcal{R}_{\Lambda}(J^{\mathrm{desc}}_{\Lambda})$$

is continuous in the Fisher topology. Define the associated branchwise descendant overlap function by

$$m_I(\Lambda) := \rho^{\mathrm{dom}}_{\Lambda}(J^{\mathrm{desc}}_{\Lambda}) = \overline{\rho}^{\mathrm{dom}}_{\Lambda}\big([\![\Xi^{(1)}_{\Lambda}]\!]\big). \tag{28}$$

Proposition 4.32 (Continuity of the descended overlap scalar)**.** *Let I be a connected accepted-branch segment. Assume:*

1. *the hypotheses of theorem 4.8 on I;*
2. *quotient descent holds for every $\Lambda \in I$;*
3. *the branchwise choice of representatives in definition 4.31 exists.*

Then the scalar function $m_I : I \to \mathbb{R}$ is well defined, independent of the chosen branchwise representatives, and continuous. Moreover,

$$|m_I(\Lambda)| = \nu_{\Lambda} \qquad \forall \Lambda \in I. \tag{29}$$

Proof. Independence of the chosen branchwise representatives is exactly the coset constancy proved in corollary 4.13: if $J'^{\mathrm{desc}}_\Lambda - J^{\mathrm{desc}}_\Lambda \in \mathcal{A}^-_\Lambda$, then

$$\rho^{\mathrm{dom}}_\Lambda(J'^{\mathrm{desc}}_\Lambda) = \rho^{\mathrm{dom}}_\Lambda(J^{\mathrm{desc}}_\Lambda) = \overline{\rho}^{\mathrm{dom}}_\Lambda\big([\![\Xi^{(1)}_\Lambda]\!]\big).$$

Thus m_I is well defined.

By theorem 4.8, the normalized dominant vector $\widehat{\delta C}^{\mathrm{LoG}}_\Lambda$ and the dominant projector $\Pi^{\mathrm{dom}}_\Lambda$ depend continuously on $\Lambda \in I$. By assumption, the response vector

$$X_\Lambda := \Pi_{\mathrm{phys},\Lambda}\mathcal{R}_\Lambda(J^{\mathrm{desc}}_\Lambda)$$

is also continuous on I. Using definition 4.17, we have

$$\Pi^{\mathrm{dom}}_\Lambda X_\Lambda = m_I(\Lambda)\,\widehat{\delta C}^{\mathrm{LoG}}_\Lambda.$$

The left-hand side is continuous because both factors are continuous, while $\widehat{\delta C}^{\mathrm{LoG}}_\Lambda \neq 0$ for all $\Lambda \in I$ by theorem 4.8. Therefore the scalar coefficient $m_I(\Lambda)$ is continuous.

Finally, (29) is just the definition of ν_Λ from definition 4.19 together with (28). □

Theorem 4.33 (Branchwise orientation transport on the descendant class)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of proposition 4.32. Then:*

1. *on every connected subsegment $I' \subset I$ such that*

$$\nu_\Lambda > 0 \qquad \forall \Lambda \in I',$$

   the sign

$$\varepsilon_\Lambda := \operatorname{sgn} m_I(\Lambda) = \operatorname{sgn}\Big(\overline{\rho}^{\mathrm{dom}}_\Lambda\big([\![\Xi^{(1)}_\Lambda]\!]\big)\Big) \in \{\pm 1\} \tag{30}$$

   is constant on I';

2. *if $\nu_{\Lambda_0} > 0$ at one point $\Lambda_0 \in I$, then there exists a neighborhood $I_0 \subset I$ of Λ_0 on which ε_Λ is constant and equal to ε_{Λ_0};*

3. *under the present hypotheses, a change of sign can occur only at a cutoff where the descendant overlap closes $\nu_\Lambda = 0$.*

In particular, once dominant-line realization, quotient descent, and nonzero overlap are established on an accepted branch segment and the projected descendant response varies continuously there, the branchwise sign statement of theorem 4.56 follows.

Proof. By proposition 4.32, $m_I(\Lambda)$ is a continuous real-valued function on I. On any connected subsegment I' where $\nu_\Lambda = |m_I(\Lambda)| > 0$, the function m_I never vanishes. A continuous nonvanishing real-valued function on a connected set has constant sign, proving item (1).

Item (2) is the local version of the same argument: if $m_I(\Lambda_0) \neq 0$, continuity gives a neighborhood I_0 on which m_I keeps the sign it has at Λ_0.

For item (3), any sign change would force m_I to pass through zero by the intermediate value theorem, hence $\nu_\Lambda = |m_I(\Lambda)| = 0$ at some intermediate cutoff. □

Corollary 4.34 (Local orientation persistence from one-tick activity)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of proposition 4.32. If the transported descendant class is dominant-active at one cutoff $\Lambda_0 \in I$, i.e.*

$$\nu_{\Lambda_0} > 0,$$

then there exists a neighborhood $I_0 \subset I$ of Λ_0 such that

$$\varepsilon_\Lambda = \varepsilon_{\Lambda_0} \qquad \forall \Lambda \in I_0.$$

Proof. This is exactly item (2) of theorem 4.33. □

Witness-indexed seam realization and scalar bridge

The bridge used below factors through a common LS witness datum rather than a direct identification of a UOM seam source with a descendant representative. On the UOM side one realizes a filler tag together with a detector witness by an admissible odd welded source, measures the resulting dominant physical pairing, and then compares that scalar with the odd representative attached to the same witness.

Definition 4.35 (Accepted record payload of a welded odd source)**.** Fix an accepted tick k. For an admissible welded odd trace-channel source τ, let

$$\rho_{\Lambda_k}[\tau]$$

denote the receiver state at cutoff Λ_k produced by that source in the welded UOM channel. Define its accepted record payload by

$$\mathcal{P}^{\mathrm{acc}}_{\Lambda_k}[\tau] := \Pi_{\Delta(\Lambda_k)}\big(\rho_{\Lambda_k}[\tau]\big). \tag{31}$$

This is the diagonal/record part retained by the finite-cutoff acceptance map.

Definition 4.36 (Witness-indexed seam realization at an accepted tick)**.** Fix an accepted tick k. Let y be a witness on the LS difference object, and write $\mathrm{flip}(y)$ for the induced LS involution. A *witness-indexed seam realization* at Λ_k is a choice of admissible welded odd trace-channel sources

$$\tau^{f,y}_{\Lambda_k}, \qquad f \in \{\mathrm{diag}, \mathrm{bridge}\},$$

such that:

1. **filler toggle changes the odd seam sign:**

$$\tau^{\mathrm{bridge},y}_{\Lambda_k} = -\,\tau^{\mathrm{diag},y}_{\Lambda_k};$$

2. **witness flip changes the odd seam sign:**

$$\tau^{f,\mathrm{flip}(y)}_{\Lambda_k} = -\,\tau^{f,y}_{\Lambda_k} \qquad \forall f \in \{\mathrm{diag}, \mathrm{bridge}\};$$

3. **accepted record payload is unchanged under filler toggle:**

$$\mathcal{P}^{\mathrm{acc}}_{\Lambda_k}\big[\tau^{\mathrm{bridge},y}_{\Lambda_k}\big] = \mathcal{P}^{\mathrm{acc}}_{\Lambda_k}\big[\tau^{\mathrm{diag},y}_{\Lambda_k}\big].$$

Remark 4.37 (Accepted-record invariance versus seam-scalar use)*.* The accepted-record invariance clause in definition 4.36 encodes the physical indistinguishability of filler toggle at the accepted-record layer. The seam-scalar theorems below use only the induced odd sign relations in the source channel.

Definition 4.38 (Raw and normalized seam pairings)**.** Fix an accepted tick k and a witness-indexed seam realization as in definition 4.36. For $f \in \{\mathrm{diag}, \mathrm{bridge}\}$ and witness y, define the *raw seam pairing* by

$$\widetilde{s}_{\Lambda_k}(f,y) := \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{f,y}_{\Lambda_k}],\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{32}$$

Whenever dominant-line realization holds so that $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ is defined, also set

$$s_{\Lambda_k}(f,y) := \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{f,y}_{\Lambda_k}],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{33}$$

Proposition 4.39 (Filler and witness parity of the seam scalar)**.** *Fix an accepted tick k and assume a witness-indexed seam realization at Λ_k. Then the raw seam pairing satisfies*

$$\widetilde{s}_{\Lambda_k}(\mathrm{bridge}, y) = -\,\widetilde{s}_{\Lambda_k}(\mathrm{diag}, y), \qquad \widetilde{s}_{\Lambda_k}(f, \mathrm{flip}(y)) = -\,\widetilde{s}_{\Lambda_k}(f, y). \tag{34}$$

If dominant-line realization also holds, then the same identities hold for the normalized seam pairing:

$$s_{\Lambda_k}(\mathrm{bridge}, y) = -\,s_{\Lambda_k}(\mathrm{diag}, y), \qquad s_{\Lambda_k}(f, \mathrm{flip}(y)) = -\,s_{\Lambda_k}(f, y). \tag{35}$$

Proof. The map $\tau \mapsto \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau]$ is linear because T_{Λ_k} and $\Pi_{\mathrm{phys},\Lambda_k}$ are linear. The Fisher pairing is bilinear. Applying the sign relations of definition 4.36 therefore gives

$$\Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{\mathrm{bridge},y}_{\Lambda_k}] = -\,\Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{\mathrm{diag},y}_{\Lambda_k}]$$

and

$$\Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{f,\mathrm{flip}(y)}_{\Lambda_k}] = -\,\Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{f,y}_{\Lambda_k}].$$

Pairing with $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ proves (34). If dominant-line realization holds, then $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ is a fixed nonzero normalization of the same vector, so the same argument proves (35). □

Definition 4.40 (Detector-factorized seam coupling)**.** Fix an accepted tick k. A witness-indexed seam realization is called *detector-factorized* if there exists a single admissible welded odd trace-channel source

$$\tau^{\mathrm{unit}}_{\Lambda_k}$$

and the LS detector factor

$$\overline{\mathrm{obs}} : \text{LS difference object} \to \mathbb{R}$$

such that, for every filler tag f and witness y,

$$\tau^{f,y}_{\Lambda_k} = \sigma_{\mathrm{fill}}(f)\, \overline{\mathrm{obs}}(y)\, \tau^{\mathrm{unit}}_{\Lambda_k}, \qquad \sigma_{\mathrm{fill}}(\mathrm{diag}) = +1, \quad \sigma_{\mathrm{fill}}(\mathrm{bridge}) = -1. \tag{36}$$

Theorem 4.41 (Seam faithfulness scalar)**.** *Fix an accepted tick k and assume a detector-factorized witness-indexed seam realization at Λ_k. Define the fixed-cutoff seam coefficient by*

$$\widetilde{c}_{\Lambda_k} := \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{\mathrm{unit}}_{\Lambda_k}],\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{37}$$

Then, for every filler tag f and witness y,

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\, \widetilde{c}_{\Lambda_k}\, \overline{\mathrm{obs}}(y). \tag{38}$$

In particular, if

$$\widetilde{c}_{\Lambda_k} \neq 0 \qquad \textit{and} \qquad \overline{\mathrm{obs}}(y) \neq 0,$$

then the raw seam pairing is nonzero:

$$\widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

Consequently the compressed welded LoG line is transversely physical at Λ_k, hence dominant-line realization holds at that cutoff by proposition 4.7. When dominant-line realization holds, the normalized seam coefficient

$$c_{\Lambda_k} := \kappa_{\Lambda_k}^{-1/2}\,\widetilde{c}_{\Lambda_k} \tag{39}$$

is defined and

$$s_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,c_{\Lambda_k}\,\overline{\mathrm{obs}}(y). \tag{40}$$

Proof. Substitute the detector-factorized form (36) into the definition (32). Linearity of T_{Λ_k} and bilinearity of the Fisher pairing give

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,\overline{\mathrm{obs}}(y)\,\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{\mathrm{unit}}\big],\,\delta C_{\Lambda_k}^{\mathrm{LoG}}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}},$$

which is exactly (38).

If $\widetilde{c}_{\Lambda_k} \neq 0$ and $\overline{\mathrm{obs}}(y) \neq 0$, then $\widetilde{s}_{\Lambda_k}(f,y) \neq 0$. Set

$$B_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{f,y}\big].$$

Then

$$\big\langle\!\big\langle \delta C_{\Lambda_k}^{\mathrm{LoG}},\,B_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} = \widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

By item (4) of proposition 4.7, this proves transversality of the compressed line and hence dominant-line realization at Λ_k.

Once dominant-line realization holds, divide (38) by $\|\delta C_{\Lambda_k}^{\mathrm{LoG}}\|_{C_{*,\Lambda_k}} = \kappa_{\Lambda_k}^{1/2}$ to obtain (40). □

Remark 4.42 (Downstream role of the seam-coefficient hypothesis). The fixed-cutoff nonvanishing hypothesis $\widetilde{c}_{\Lambda_k} \neq 0$ is exactly the input that, once dominant-line realization defines c_{Λ_k}, is imported downstream as the nondegeneracy clause $c_{\Lambda_k} \neq 0$ inside the delayed selected-line package. This note does not derive that clause from split-threshold analysis alone.

Definition 4.43 (Odd source representability on the characterization layer)**.** Fix an accepted tick k. Let τ be an admissible welded odd trace-channel source. We say that τ is *representable on the characterization layer* if there exists an odd sufficiently local generator

$$F_{\tau,\Lambda_k} \in E_{\Lambda_k}^{-}$$

such that its intrinsic physical covariance response agrees with the UOM welded response:

$$\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big) = \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau]. \tag{41}$$

This is the strong source-insertion realization statement.

Theorem 4.44 (UOM source-insertion representability theorem)**.** *Fix an accepted tick k. Let τ be an admissible welded odd trace-channel source at cutoff Λ_k, in the finite-cutoff UOM sense: its receiver-head profile is band-limited at Λ_k and compactly supported inside the accepted slab. Then there exists a canonically associated odd sufficiently local generator*

$$F_{\tau,\Lambda_k} \in E_{\Lambda_k}^{-}$$

such that

$$\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big) = \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau]. \tag{42}$$

In particular, every witness-indexed seam source used in the present note is representable on the characterization layer.

Proof. By the finite-cutoff UOM Peierls package, admissible receiver-head sources at cutoff Λ_k are band-limited boundary test data and the sufficiently local class $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}$ contains all compact smearings of local boundary densities. Let

$$F_{\tau,\Lambda_k}$$

be the linear source functional obtained by pairing the odd trace-channel source profile τ with the corresponding odd trace-channel boundary field on the receiver head. Because this is the linear case of the sufficiently local form and τ is physically odd,

$$F_{\tau,\Lambda_k} \in E^-_{\Lambda_k}.$$

Now apply the UOM source insertion map from the finite-cutoff retarded-kernel package. The source τ is sent to the bulk forcing term in the linearized equation; the unique retarded solution is obtained by the retarded Green operator; and boundary restriction gives the retarded covariance variation. By the UOM identification of the induced retarded kernel with the SK mixed second derivative, this covariance variation is exactly the welded source-to-covariance response

$$\mathsf{T}_{\Lambda_k}[\tau].$$

On the other hand, by the definition of the intrinsic covariance-response map, the matrix elements of $\mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k})$ are computed by the Peierls action of F_{τ,Λ_k} on the sufficiently local quadratic probes representing boundary covariance entries. That Peierls variation is precisely the same retarded second-moment variation generated by the source insertion τ. Therefore

$$\mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k}) = \mathsf{T}_{\Lambda_k}[\tau]$$

as covariance tangents. Applying the physical projector $\Pi_{\mathrm{phys},\Lambda_k}$ gives (42).

The witness-indexed seam sources of definition 4.36 are, by construction, live admissible welded odd trace-channel sources at the accepted tick, so the same argument applies to each of them. □

Proposition 4.45 (Dominant pairing for a representable odd source)**.** *Fix an accepted tick k and let τ be an admissible welded odd trace-channel source that is representable in the sense of theorem 4.44. Then*

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau],\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{43}$$

If dominant-line realization also holds, then

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big) = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{44}$$

Proof. Equation (41) identifies the two physical covariance tangents. Pairing both sides with $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ gives (43). If dominant-line realization holds, pair with the normalized vector $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ instead and use the definition (22) of $\rho^{\mathrm{dom}}_{\Lambda_k}$ to obtain (44). □

Corollary 4.46 (Unit welded odd source representability and scalar identification)**.** *Fix an accepted tick k. Assume a detector-factorized witness-indexed seam realization at Λ_k in the sense of definition 4.40, and assume dominant-line realization at Λ_k. Then the unit welded odd source $\tau^{\mathrm{unit}}_{\Lambda_k}$ is representable on the characterization layer, with canonical representable generator*

$$F_{\tau^{\mathrm{unit}}_{\Lambda_k},\Lambda_k} \in E^-_{\Lambda_k},$$

and

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{\mathrm{unit}}_{\Lambda_k},\Lambda_k}\big) = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{unit}}_{\Lambda_k}\big],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = c_{\Lambda_k}. \tag{45}$$

Proof. Representability of $\tau^{\mathrm{unit}}_{\Lambda_k}$ is the specialization of theorem 4.44 to the admissible welded odd trace source singled out by definition 4.40. Because dominant-line realization holds, proposition 4.45 applies to that representable source and gives the first equality in (45). The second equality is exactly the definition of the normalized seam coefficient c_{Λ_k} from theorem 4.41. □

Definition 4.47 (Scalar comparison shadow of the witness-indexed bridge)**.** Fix an accepted tick k. Let y be an LS witness and let $\tau^{f,y}_{\Lambda_k}$ be a witness-indexed seam realization source for a filler tag f. A *scalar comparison shadow* for that data is a choice of representative

$$J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

such that the dominant raw pairing of the 2T-side representative agrees with the dominant raw pairing of the corresponding representable UOM source:

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{46}$$

When dominant-line realization holds, this is equivalently the equality of normalized dominant pairings

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big). \tag{47}$$

Definition 4.48 (Witness-indexed source-compatible descendant realization)**.** Fix an accepted tick k, a filler tag f, a witness y, and a witness-indexed seam source $\tau^{f,y}_{\Lambda_k}$. A representative

$$J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

is called a *witness-indexed source-compatible descendant realization* if it realizes, on the same LS witness datum, the Limited Scope output and odd Hamiltonization clauses from item **(M1)** and item **(M4)** of Definition 4.2 and differs from the source-insertion representative only by an ambiguity direction:

$$J^{f,y}_{\Lambda_k} - F_{\tau^{f,y}_{\Lambda_k},\Lambda_k} \in \mathcal{A}^{-}_{\Lambda_k}, \tag{48}$$

where $F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}$ is the representable source generator furnished by theorem 4.44. Equivalently, the witness-indexed transported descendant class and the witness-indexed UOM source-insertion realization define the same point of the ambiguity quotient $E^{-}_{\Lambda_k}/\mathcal{A}^{-}_{\Lambda_k}$.

Theorem 4.49 (Same-witness descended-scalar comparison from a source-compatible descendant realization)**.** *Fix an accepted tick k, assume dominant-line realization at Λ_k, and let $\tau^{f,y}_{\Lambda_k}$ be a witness-indexed seam realization source. Assume:*

1. *the source $\tau^{f,y}_{\Lambda_k}$ is represented by the canonical generator $F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}$ from theorem 4.44;*
2. *there exists a witness-indexed source-compatible descendant realization $J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ in the sense of definition 4.48;*
3. *the three vanishing statements of proposition 4.15 hold at Λ_k.*

Then the raw and normalized comparison identities of definition 4.47 hold:

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}, \tag{49}$$

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big). \tag{50}$$

Moreover,

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big) = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{f,y}_{\Lambda_k}\big],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{51}$$

If quotient descent also holds at Λ_k, then the common value equals the descended dominant scalar on the transported descendant affine class:

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big). \tag{52}$$

Proof. Set

$$A^{f,y}_{\Lambda_k} := J^{f,y}_{\Lambda_k} - F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}.$$

By source compatibility, $A^{f,y}_{\Lambda_k} \in \mathcal{A}^{-}_{\Lambda_k}$.

Because dominant-line realization holds and the three vanishing statements of proposition 4.15 are assumed, that proposition gives

$$\mathcal{A}^{-}_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k}.$$

Hence

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k} + A^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),$$

proving (50).

Since dominant-line realization implies $\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0$ and $\rho^{\mathrm{dom}}_{\Lambda_k} = \kappa^{-1/2}_{\Lambda_k}\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$, the raw pairing equality (49) is just the same identity multiplied by the positive factor $\kappa^{1/2}_{\Lambda_k}$.

Equation (51) is exactly the normalized pairing formula of proposition 4.45 applied to the representable source $\tau^{f,y}_{\Lambda_k}$.

If quotient descent also holds, then corollary 4.13 gives

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)$$

for every representative $J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$, which yields (52) after combining with (50). □

Theorem 4.50 (Fixed-cutoff seam witness criterion for descendant overlap)**.** *Fix an accepted tick k. Assume:*

1. *a detector-factorized witness-indexed seam realization at Λ_k;*
2. *a witness y with*
$$\overline{\mathrm{obs}}(y) \neq 0;$$
3. *the seam coefficient is nonzero,*
$$\widetilde{c}_{\Lambda_k} \neq 0;$$
4. *for the chosen filler tag f, the source $\tau^{f,y}_{\Lambda_k}$ is representable on the characterization layer in the sense of definition 4.43;*
5. *there exists a witness-indexed source-compatible descendant realization in the sense of definition 4.48.*

Then:

1. *the compressed welded LoG line is transversely physical at Λ_k, hence dominant-line realization holds at that cutoff;*

2. *the chosen 2T representative satisfies*

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) \neq 0;$$

3. *therefore the transported descendant class has nonzero overlap with the canonically selected compressed dominant odd line at* Λ_k*;*

4. *if quotient descent also holds at* Λ_k*, then theorem 4.21 applies and the transported descendant class is dominant-active with well-defined sign* ε_{Λ_k}*.*

Proof. By theorem 4.41,

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,\widetilde{c}_{\Lambda_k}\,\overline{\mathrm{obs}}(y) \neq 0.$$

The same theorem therefore gives item (1): the compressed line is transversely physical, hence dominant-line realization holds at Λ_k.

By proposition 4.45,

$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} = \widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

Applying the comparison identity (46) now gives

$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} \neq 0.$$

Because dominant-line realization has already been established, $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ is nonzero and $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ is defined. By proposition 4.18, this is equivalent to $\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) \neq 0$, proving item (2).

Item (3) is exactly the nonzero-overlap condition from proposition 4.20. If quotient descent is also available, apply theorem 4.21 to obtain item (4). □

Smooth-taper branch continuity and witness transport

The previous fixed-cutoff theorem isolates the nonzero witness needed for activity. To transport that witness along an accepted branch one needs a continuity statement for the welded response. At theorem level, the clean analytic choice is the smooth receiver taper. The hard projector may still be used, but only on branch segments where its rank is constant.

Lemma 4.51 (Smooth-taper continuity of the receiver band projector)**.** *Let* I *be a connected accepted-branch segment. Suppose the receiver band is implemented by a smooth taper*

$$P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda) = \sum_{\ell,m} w_\ell(\Lambda)\,|Y_{\ell m}\rangle\langle Y_{\ell m}|, \qquad 0 \le w_\ell(\Lambda) \le 1,$$

on the receiver one-particle space, where

$$\sup_{\ell\ge 0} |w_\ell(\Lambda') - w_\ell(\Lambda)| \longrightarrow 0 \qquad \text{as } \Lambda' \to \Lambda$$

for every $\Lambda \in I$*. Then*

$$\Lambda \longmapsto P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda)$$

is continuous on I *in operator norm. If instead the hard projector* $\sum_{\ell\le\ell_\star(\Lambda)} |Y_{\ell m}\rangle\langle Y_{\ell m}|$ *is used, the same conclusion holds on every connected subsegment of* I *on which* $\ell_\star(\Lambda)$ *is constant.*

Proof. In the spherical-harmonic basis $\{Y_{\ell m}\}$, the difference

$$P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda') - P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda)$$

is diagonal with eigenvalues $w_\ell(\Lambda') - w_\ell(\Lambda)$. Therefore

$$\left\| P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda') - P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda) \right\|_{\mathrm{op}} = \sup_{\ell \geq 0} |w_\ell(\Lambda') - w_\ell(\Lambda)|.$$

The assumed uniform continuity of the taper coefficients is exactly the statement that this operator norm tends to zero as $\Lambda' \to \Lambda$.

For the hard projector, the coefficients take only the values 0 and 1, so discontinuities can occur only when $\ell_\star(\Lambda)$ jumps. On a subsegment where $\ell_\star(\Lambda)$ is constant, the projector is constant and therefore continuous. □

Theorem 4.52 (Smooth-taper branch continuity of the welded physical response)**.** *Let I be a connected accepted-branch segment. Write the welded source-to-covariance map in the UOM factorized form*

$$\mathsf{T}_\Lambda = \Pi_{L(\Lambda)} M_{\varepsilon,\Lambda} \circ \mathsf{K}_{\mathrm{comp},\Lambda} \circ \mathsf{G}_{\mathrm{ret},\Lambda} \circ \mathsf{W}_{\mathrm{src},\Lambda}. \tag{53}$$

Assume:

1. *the receiver band is implemented by a smooth taper satisfying lemma 4.51;*
2. *the operator families*

$$\Lambda \longmapsto \Pi_{L(\Lambda)} M_{\varepsilon,\Lambda}, \qquad \Lambda \longmapsto \mathsf{K}_{\mathrm{comp},\Lambda}, \qquad \Lambda \longmapsto \mathsf{G}_{\mathrm{ret},\Lambda}, \qquad \Lambda \longmapsto \mathsf{W}_{\mathrm{src},\Lambda}, \qquad \Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda}$$

   are continuous on I in operator norm on the corresponding band-limited source and covariance tangent spaces;

3. *the source family*

$$\Lambda \longmapsto \tau_\Lambda$$

   is continuous on I in the welded source norm.

Then the physical covariance response

$$X_\Lambda[\tau_\Lambda] := \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda[\tau_\Lambda] \tag{54}$$

depends continuously on $\Lambda \in I$ in the Fisher norm. If the hard receiver projector is used instead, the same conclusion holds on each connected subsegment where $\ell_\star(\Lambda)$ is constant.

Proof. By lemma 4.51, the first factor $\Pi_{L(\Lambda)} M_{\varepsilon,\Lambda}$ is operator-norm continuous along the branch. By assumption, the remaining three UOM factors in (53) and the physical projector $\Pi_{\mathrm{phys},\Lambda}$ are also operator-norm continuous. Hence the composed operator family

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda$$

is continuous in operator norm on I. Applying this continuous family to the continuous source vector τ_Λ gives continuity of $X_\Lambda[\tau_\Lambda]$ in the Fisher norm.

For the hard projector, the same argument applies on every connected subsegment where $\ell_\star(\Lambda)$ is constant, because there the first factor is constant. □

Corollary 4.53 (Branchwise continuity of the seam coefficient)**.** *Let I be a connected accepted-branch segment and assume:*

1. *the hypotheses of theorem 4.8 on I;*
2. *a detector-factorized branchwise seam realization on I with continuous unit source*

$$\Lambda \longmapsto \tau_\Lambda^{\mathrm{unit}};$$

3. *the hypotheses of theorem 4.52 for that unit source.*

Define the branchwise normalized seam coefficient by

$$c_\Lambda := \langle\!\langle \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda\big[\tau_\Lambda^{\mathrm{unit}}\big],\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}} \rangle\!\rangle_{C_{*,\Lambda}}. \tag{55}$$

Then c_Λ is continuous on I. Consequently, for every fixed filler tag f and witness y, the normalized seam scalar

$$s_\Lambda(f,y) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y)$$

is continuous on I.

Proof. By theorem 4.52, the response vector

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda\big[\tau_\Lambda^{\mathrm{unit}}\big]$$

is continuous in the Fisher norm. By theorem 4.8, the normalized dominant vector $\widehat{\delta C}_\Lambda^{\mathrm{LoG}}$ is also continuous on I. Pairing these two continuous vectors in the Fisher metric gives continuity of c_Λ. The final statement is then immediate from (40). □

Theorem 4.54 (Branchwise seam witness transport)**.** *Let I be a connected accepted-branch segment. Assume:*

1. *the hypotheses of corollary 4.53;*
2. *quotient descent holds for every $\Lambda \in I$;*
3. *there exists a fixed LS witness y with*

$$\overline{\mathrm{obs}}(y) \neq 0;$$

4. *for some fixed filler tag f, there is a branchwise family of representatives*

$$J_\Lambda^{f,y} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

such that, for every $\Lambda \in I$, the corresponding seam source $\tau_\Lambda^{f,y}$ is representable on the characterization layer and the 2T-side witness-to-descendant comparison identity (47) holds.

Then:

1. *the branchwise dominant-response scalar of the chosen descendant representatives is*

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y); \tag{56}$$

2. *the scalar function* $\Lambda \mapsto \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})$ *is continuous on* I*;*

3. *on every connected subsegment* $I' \subset I$ *on which* $c_\Lambda \neq 0$*, the sign*

$$\operatorname{sgn}\big(\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})\big)$$

is constant;

4. *because quotient descent holds, the same constant sign is the transported orientation sign of the descendant affine class on* I'*.*

Proof. For each $\Lambda \in I$, representability of $\tau_\Lambda^{f,y}$ and proposition 4.45 give

$$\rho_\Lambda^{\mathrm{dom}}\big(F_{\tau_\Lambda^{f,y},\Lambda}\big) = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda\big[\tau_\Lambda^{f,y}\big],\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}} \big\rangle\!\big\rangle_{C_{*,\Lambda}} = s_\Lambda(f,y).$$

Applying the branchwise comparison identity (47) then yields

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = s_\Lambda(f,y).$$

Now use corollary 4.53 to substitute $s_\Lambda(f,y) = \sigma_{\mathrm{fill}}(f) c_\Lambda \overline{\mathrm{obs}}(y)$, which proves item (1). Item (2) follows because c_Λ is continuous and $\overline{\mathrm{obs}}(y)$ is a fixed nonzero scalar. On any connected subsegment where $c_\Lambda \neq 0$, the right-hand side of (56) is a continuous nonvanishing real-valued function, so its sign is constant. This proves item (3).

Finally, quotient descent implies that $\rho_\Lambda^{\mathrm{dom}}$ is constant on $[\Xi_\Lambda^{(1)}]$, so the sign of one representative is the sign of the whole descendant class. This gives item (4). □

Corollary 4.55 (Seam package discharge of the minimal scalar transport input)**.** *Let* I *be a connected accepted-branch segment and assume the hypotheses of theorem 4.54. Then the branchwise family*

$$J_\Lambda^{f,y} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

satisfies

$$\Lambda \longmapsto \rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y)$$

continuously on I*. On every connected subsegment* $I' \subset I$ *such that*

$$c_\Lambda \neq 0 \qquad \forall \Lambda \in I',$$

assumption (3) *of theorem 4.56 holds in its minimal scalar form with*

$$J_\Lambda^{\mathrm{desc}} := J_\Lambda^{f,y} \qquad (\Lambda \in I').$$

If, in addition, the three generator-wise vanishing statements of proposition 4.15 hold for every $\Lambda \in I'$*, then theorem 4.56 applies on* I'*.*

Proof. Continuity of the displayed scalar function is exactly item (2) of theorem 4.54, together with the explicit formula of item (1). On I', the hypotheses $c_\Lambda \neq 0$ and $\overline{\mathrm{obs}}(y) \neq 0$ imply

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) \neq 0 \qquad \forall \Lambda \in I'.$$

Together with the continuity statement already proved above, this is exactly assumption (3) of theorem 4.56 in its minimal scalar form. The hypotheses of theorem 4.54 already include those of corollary 4.53, and hence the dominant-line persistence input of theorem 4.56. If the three generator-wise vanishing statements also hold on I', then every hypothesis of theorem 4.56 is in force there. □

Theorem 4.56 (Finite-cutoff Active Response Transport Theorem)**.** *Let I be a connected accepted-branch segment. For each $\Lambda \in I$, let*

$$[\Xi_\Lambda^{(1)}] := J_{0,\Lambda} + \mathcal{A}_\Lambda^- \subset E_\Lambda^-$$

be the parity-transported odd descendant affine class modulo the ambiguity space of definition 4.2. Assume:

1. *the hypotheses of theorem 4.8 hold on I;*
2. *for every $\Lambda \in I$, the three vanishing statements of proposition 4.15 hold;*
3. *there exists a branchwise family of representatives*

$$J_\Lambda^{\mathrm{desc}} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

such that

$$\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) \neq 0 \qquad \forall \Lambda \in I,$$

and such that the dominant-response scalar

$$\Lambda \longmapsto \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}})$$

is continuous on I.

Then, for every $\Lambda \in I$:

1. *the compressed dominant covariance tangent is nonzero,*

$$\delta C_\Lambda^{\mathrm{LoG}} \neq 0,$$

so the dominant line L_Λ^{dom} of definition 4.3 is canonically defined;

2. *the dominant-response coefficient $\rho_\Lambda^{\mathrm{dom}}$ annihilates the odd ambiguity space,*

$$\rho_\Lambda^{\mathrm{dom}}(A) = 0 \qquad \forall A \in \mathcal{A}_\Lambda^-;$$

3. *the transported descendant affine class carries a unique nonzero dominant-response value*

$$\rho_\Lambda^{\mathrm{dom}}(J) = \overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big) \neq 0 \qquad \forall J \in [\Xi_\Lambda^{(1)}],$$

and hence a well-defined sign

$$\varepsilon_\Lambda := \mathrm{sgn}\Big(\overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big)\Big) \in \{\pm 1\};$$

4. *the normalized orientation functional*

$$\chi_\Lambda := \nu_\Lambda^{-1} \rho_\Lambda^{\mathrm{dom}} \tag{57}$$

is well defined, annihilates $\mathcal{A}_\Lambda^-$, and satisfies

$$\chi_\Lambda(J) = \varepsilon_\Lambda \qquad \forall J \in [\Xi_\Lambda^{(1)}];$$

5. *along I, the sign ε_Λ is locally constant once an initial orientation is fixed.*

Proof. Item (1) is exactly the conclusion of theorem 4.8.

By assumption, the three generator-wise vanishing statements of proposition 4.15 hold for every $\Lambda \in I$. Therefore proposition 4.15 and theorem 4.16 give item (2) and the existence of the descended functional $\overline{\rho}^{\mathrm{dom}}_\Lambda$ at each cutoff.

Applying proposition 4.20 and theorem 4.21 pointwise to the nonvanishing scalar assumption $\rho^{\mathrm{dom}}_\Lambda(J^{\mathrm{desc}}_\Lambda) \neq 0$ now yields item (3): the entire descendant affine class has one common nonzero dominant-response value, and its sign ε_Λ is well defined.

Since $\nu_\Lambda > 0$ by item (3), the normalized functional χ_Λ of (57) is well defined. Item (2) implies that χ_Λ annihilates $\mathcal{A}^-_\Lambda$, and item (3) gives

$$\chi_\Lambda(J) = \frac{\rho^{\mathrm{dom}}_\Lambda(J)}{\left|\overline{\rho}^{\mathrm{dom}}_\Lambda\big([\![\Xi^{(1)}_\Lambda]\!]\big)\right|} = \varepsilon_\Lambda \qquad \forall J \in [\Xi^{(1)}_\Lambda],$$

proving item (4).

Finally, assumption (3) provides a continuous nonvanishing real-valued branchwise descendant scalar, while items (1)–(3) above identify that scalar with the transported orientation sign of the full descendant affine class. Since $\Lambda \mapsto \rho^{\mathrm{dom}}_\Lambda(J^{\mathrm{desc}}_\Lambda)$ is continuous and nonzero on the connected set I, its sign is constant, hence in particular locally constant. Using item (3), this is exactly the transported orientation sign of the full descendant affine class, proving item (5). □

Proposition 4.57 (Ambiguity-triviality on the compressed welded image)**.** *Assume the hypotheses of theorem 4.56 at an accepted tick k. Define the reduced welded image of an odd generator $F \in E^-_{\Lambda_k}$ by*

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F) := \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, P^{\mathrm{LoG}}_{-,\Lambda_k},$$

where $P^{\mathrm{LoG}}_{-,\Lambda_k}$ is the canonical compressed odd projector of corollary 1.10. Then

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F + A) = \mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F) \qquad \forall F \in E^-_{\Lambda_k},\ \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Equivalently, $\mathsf{W}^{\mathrm{red}}_{\Lambda_k}$ factors through the ambiguity quotient $E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$. In particular, if $J_1, J_2 \in [\Xi^{(1)}_{\Lambda_k}]$, then

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(J_1) = \mathsf{W}^{\mathrm{red}}_{\Lambda_k}(J_2).$$

Proof. Item (2) of theorem 4.56 gives

$$\rho^{\mathrm{dom}}_{\Lambda_k}(A) = 0 \qquad \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Therefore, for every $F \in E^-_{\Lambda_k}$ and $A \in \mathcal{A}^-_{\Lambda_k}$,

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F + A) = \rho^{\mathrm{dom}}_{\Lambda_k}(F + A)\, P^{\mathrm{LoG}}_{-,\Lambda_k} = \big(\rho^{\mathrm{dom}}_{\Lambda_k}(F) + \rho^{\mathrm{dom}}_{\Lambda_k}(A)\big) P^{\mathrm{LoG}}_{-,\Lambda_k} = \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, P^{\mathrm{LoG}}_{-,\Lambda_k}.$$

Since corollary 1.10 fixes $P^{\mathrm{LoG}}_{-,\Lambda_k}$ canonically on the compressed, co-centered, band-locked welded channel, the right-hand side depends only on the ambiguity class of F. If $J_2 - J_1 \in \mathcal{A}^-_{\Lambda_k}$, apply the same identity with $F = J_1$ and $A = J_2 - J_1$. □

Proposition 4.58 (Construction of an admissible calibration for the minimal transported interface)**.** *Fix an accepted tick k. Suppose a weak transported-interface package is given:*

$$\mathcal{A}^-_{\Lambda_k} \subset E^-_{\Lambda_k}, \qquad [\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^-_{\Lambda_k},$$

together with a continuous nonzero linear functional

$$\chi_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

and a sign $\varepsilon_{\Lambda_k} \in \{\pm 1\}$ *such that*

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \chi_{\Lambda_k}, \qquad \chi_{\Lambda_k}(J) = \varepsilon_{\Lambda_k} \quad \forall J \in [\Xi^{(1)}_{\Lambda_k}].$$

Assume in addition that $\ker \chi_{\Lambda_k}$ *admits a continuous symmetric coercive bilinear form whose induced norm Hilbertizes it. Choose any vector* $J^\sharp_{\Lambda_k} \in E^-_{\Lambda_k}$ *satisfying*

$$\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1,$$

any scale $\beta_{\Lambda_k} > 0$*, and any continuous symmetric coercive bilinear form*

$$\langle \cdot, \cdot \rangle^{\mathrm{aux},\chi}_{\Lambda_k} : \ker \chi_{\Lambda_k} \times \ker \chi_{\Lambda_k} \to \mathbb{R}$$

whose induced norm Hilbertizes $\ker \chi_{\Lambda_k}$*. Define*

$$\mathsf{B}_{\Lambda_k}(F, G) := \beta_{\Lambda_k}\, \chi_{\Lambda_k}(F)\chi_{\Lambda_k}(G) + \Big\langle F - \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k},\, G - \chi_{\Lambda_k}(G) J^\sharp_{\Lambda_k} \Big\rangle^{\mathrm{aux},\chi}_{\Lambda_k}. \tag{58}$$

Then B_{Λ_k} *is a continuous symmetric positive definite bilinear form on* $E^-_{\Lambda_k}$ *whose induced norm Hilbertizes* $E^-_{\Lambda_k}$*, and*

$$\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0 \qquad \forall K \in \ker \chi_{\Lambda_k}.$$

Hence $\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0$ *in particular for every* $K \in \mathcal{A}^-_{\Lambda_k}$*, and*

$$\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, J^\sharp_{\Lambda_k}) = \beta_{\Lambda_k}.$$

Moreover, for any representative $J^\Xi_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$*, the choice*

$$J^\sharp_{\Lambda_k} := \varepsilon_{\Lambda_k} J^\Xi_{\Lambda_k}$$

is admissible.

Proof. Because $\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1$, the linear map

$$P^\chi_{\Lambda_k}(F) := F - \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k}$$

takes values in $\ker \chi_{\Lambda_k}$:

$$\chi_{\Lambda_k}\big(P^\chi_{\Lambda_k}(F)\big) = \chi_{\Lambda_k}(F) - \chi_{\Lambda_k}(F)\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 0.$$

Thus every $F \in E^-_{\Lambda_k}$ has the decomposition

$$F = \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k} + P^\chi_{\Lambda_k}(F), \qquad P^\chi_{\Lambda_k}(F) \in \ker \chi_{\Lambda_k}.$$

This decomposition is unique: if $F = aJ^\sharp_{\Lambda_k} + K$ with $K \in \ker \chi_{\Lambda_k}$, applying χ_{Λ_k} gives $a = \chi_{\Lambda_k}(F)$.

Equation (58) is therefore well defined, continuous, and symmetric. If $\mathsf{B}_{\Lambda_k}(F, F) = 0$, then

$$\beta_{\Lambda_k} \chi_{\Lambda_k}(F)^2 = 0$$

and

$$\left\langle P^{\chi}_{\Lambda_k}(F), P^{\chi}_{\Lambda_k}(F)\right\rangle^{\mathrm{aux},\chi}_{\Lambda_k} = 0.$$

Since $\beta_{\Lambda_k} > 0$, we get $\chi_{\Lambda_k}(F) = 0$, hence $P^{\chi}_{\Lambda_k}(F) = F$. Coercivity on $\ker \chi_{\Lambda_k}$ then implies $F = 0$, proving positive definiteness.

The map

$$F \longmapsto \big(\chi_{\Lambda_k}(F), P^{\chi}_{\Lambda_k}(F)\big)$$

is a continuous linear isomorphism from $E^-_{\Lambda_k}$ onto $\mathbb{R} \oplus \ker \chi_{\Lambda_k}$, with inverse

$$(a, K) \longmapsto aJ^{\sharp}_{\Lambda_k} + K.$$

Under this isomorphism, B_{Λ_k} becomes the product Hilbert form

$$(a, K), (b, L) \longmapsto \beta_{\Lambda_k} ab + \langle K, L\rangle^{\mathrm{aux},\chi}_{\Lambda_k},$$

so its induced norm Hilbertizes $E^-_{\Lambda_k}$.

Finally, if $K \in \ker \chi_{\Lambda_k}$, then $P^{\chi}_{\Lambda_k}(J^{\sharp}_{\Lambda_k}) = 0$ and $\chi_{\Lambda_k}(K) = 0$, hence

$$\mathsf{B}_{\Lambda_k}(J^{\sharp}_{\Lambda_k}, K) = 0.$$

Setting $K = J^{\sharp}_{\Lambda_k}$ gives $\mathsf{B}_{\Lambda_k}(J^{\sharp}_{\Lambda_k}, J^{\sharp}_{\Lambda_k}) = \beta_{\Lambda_k}$. If $J^{\Xi}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$, then $\chi_{\Lambda_k}(J^{\Xi}_{\Lambda_k}) = \varepsilon_{\Lambda_k}$, so $J^{\sharp}_{\Lambda_k} := \varepsilon_{\Lambda_k} J^{\Xi}_{\Lambda_k}$ satisfies $\chi_{\Lambda_k}(J^{\sharp}_{\Lambda_k}) = 1$. □

Remark 4.59 (Calibration existence versus stiffness matching). Proposition 4.58 is a construction theorem, not an unconditional existence theorem: it extends a previously chosen Hilbertizing form on $\ker \chi_{\Lambda_k}$ to an admissible bilinear form on the whole weak transported interface. It does *not* identify the scale β_{Λ_k} with the split/leak stiffness $\widetilde{\alpha}(\Lambda_k)$. That physical identification belongs to a later strong or physical layer.

Theorem 4.60 (Rank-one descendant-sector upgrade)**.** *Assume theorem 4.56, and fix an accepted tick k on the branch. Suppose in addition that on the transported descendant sector the projected response map*

$$F \longmapsto \Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F), \qquad \Pi^{\mathrm{dom}}_{\Lambda_k} := \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \otimes \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k},$$

has kernel exactly $\mathcal{A}^-_{\Lambda_k}$. Then the stronger conclusions follow:

1. *the descendant quotient is rank one,*

$$E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k} \cong \mathbb{R};$$

2. *the orientation kernel is exactly the ambiguity space,*

$$\ker \chi_{\Lambda_k} = \ker \rho^{\mathrm{dom}}_{\Lambda_k} = \mathcal{A}^-_{\Lambda_k},$$

   so the descendant sector satisfies the rank-one kernel-identification package with quotient generated by any vector $J^{\mathrm{act}}_{\Lambda_k}$ satisfying $\chi_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1$;

3. *after choosing any such $J^{\mathrm{act}}_{\Lambda_k}$, any positive scale β_{Λ_k}, and any auxiliary coercive Hilbertizing form on $\ker \chi_{\Lambda_k}$, proposition 4.58 yields an admissible calibration form. Recovering the stiffness-matched formula of proposition 3.17 requires additional physical identifications, namely $\chi_{\Lambda_k} = \rho_{\Lambda_k}$, $\ker \chi_{\Lambda_k} = \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$, agreement of the chosen normalized generator with the split/leak active generator, and $\beta_{\Lambda_k} = \widetilde{\alpha}(\Lambda_k)$.*

Proof. By theorem 4.56, the dominant line is defined at Λ_k, the scalar ν_{Λ_k} is positive, and

$$\chi_{\Lambda_k} = \nu_{\Lambda_k}^{-1} \rho_{\Lambda_k}^{\mathrm{dom}}.$$

Hence $\ker \chi_{\Lambda_k} = \ker \rho_{\Lambda_k}^{\mathrm{dom}}$.

For every $F \in E_{\Lambda_k}^{-}$, the definition of $\rho_{\Lambda_k}^{\mathrm{dom}}$ gives

$$\Pi_{\Lambda_k}^{\mathrm{dom}} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F) = \rho_{\Lambda_k}^{\mathrm{dom}}(F)\, \widehat{\delta C}_{\Lambda_k}^{\mathrm{LoG}}.$$

Therefore the kernel of the projected response map is exactly $\ker \rho_{\Lambda_k}^{\mathrm{dom}}$. By the additional hypothesis, that kernel is $\mathcal{A}_{\Lambda_k}^{-}$, proving item (2).

By theorem 4.56, there exists $J_{\Lambda_k}^{\mathrm{desc}} \in [\Xi_{\Lambda_k}^{(1)}]$ with $\rho_{\Lambda_k}^{\mathrm{dom}}(J_{\Lambda_k}^{\mathrm{desc}}) \neq 0$, hence the projected response map is nonzero. Its image is automatically contained in the one-dimensional line $L_{\Lambda_k}^{\mathrm{dom}}$, so the first isomorphism theorem gives

$$E_{\Lambda_k}^{-} / \mathcal{A}_{\Lambda_k}^{-} \cong \mathrm{im}\big(\Pi_{\Lambda_k}^{\mathrm{dom}} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}\big) \cong \mathbb{R},$$

which is item (1). Any vector $J_{\Lambda_k}^{\mathrm{act}}$ with $\chi_{\Lambda_k}(J_{\Lambda_k}^{\mathrm{act}}) = 1$ lies outside $\ker \chi_{\Lambda_k} = \mathcal{A}_{\Lambda_k}^{-}$, so its quotient class generates this one-dimensional quotient, completing item (2).

Item (3) is now an application of proposition 4.58 with the weak-interface data supplied by theorem 4.56 and the chosen normalized vector $J_{\Lambda_k}^{\mathrm{act}}$. The final sentence identifies when this weak calibration construction coincides with the stiffness-matched formula of proposition 3.17. □